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Type 2 diabetes and attenuated weight loss during incretin-based therapy: mechanisms linking glucose homeostasis to energy balance

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ABSTRACT

Clinical trials of incretin-based therapies consistently report smaller reductions in body weight in people with type 2 diabetes (T2D) than in people without diabetes. We asked how robust that observation is and which mechanisms can account for it. We identified nine comparisons in which the same molecule, dose, treatment duration and estimand were tested in separate populations with and without T2D: semaglutide (three programmes, including semaglutide 7.2 mg in STEP UP and STEP UP T2D), tirzepatide (two doses), orforglipron (two doses), cagrilintide–semaglutide and maridebart cafraglutide. In these comparisons the placebo-adjusted weight reduction in T2D was a median 0.63 (range 0.50–0.78) of that in populations without diabetes, an absolute difference of 2–7 percentage points. A prespecified pooled analysis of harmonised semaglutide heart-failure trials confirmed strong heterogeneity between the diabetes and non-diabetes populations ($I^2 \approx 96\%$). The deficit is graded across normoglycaemia, prediabetes and T2D, and it appears across agents with different receptor profiles. Placebo-arm weight change is not smaller in T2D, which places the deficit in the drug-attributable effect. Human studies show that incretin agonists suppress appetite in T2D, but no study has compared the magnitude of that suppression with that in people without diabetes. The peripheral apparatus around the receptor — endogenous secretion, degradation, and vagal afferent transmission — is measurably altered in T2D, and intact vagal innervation is required for native GLP-1 to suppress intake in humans; but incretin secretion is on average preserved and falls only with worsening glycaemic control, the portal afferent receptor appears not to be engaged by exogenous agonists, and prolonging native GLP-1 pharmacologically does not reduce weight. The deficit is also not specific to incretin drugs: it appears with pharmacotherapies acting on unrelated targets and after bariatric surgery. That places the mechanism in the host rather than in incretin receptor pharmacology, but it does not identify which host mechanism, because three remain compatible with all the evidence and are not mutually exclusive: the energetics of glycaemic correction; a generalised deficit in anorectic responsiveness; and degradation of the shared multi-axis gut–brain afferent pathway. Correction of hyperglycaemia removes urinary glucose energy losses and lowers resting energy expenditure; our calculations suggest these offsets could account for roughly 1–6 kg by 68–72 weeks, and an energy-balance model partially reproduces the gap from these terms alone, though it held appetite parameters fixed and so could not express the competing hypotheses. Neither component has been measured in an incretin trial. The deficit already present in prediabetes, where glycosuria is negligible, favours the appetite and afferent accounts; the persistence of the gap under enforced fixed intake favours the energetic one. Co-prescribed weight-promoting drugs, differences in age and sex, and behavioural factors probably contribute. Evidence for an intrinsic, insulin-resistance-driven resistance to incretin pharmacology is weak. The most important missing evidence is an energy-balance study, under matched conditions, that measures intake, expenditure and urinary energy losses in people with and without T2D receiving the same agent; energy intake is the measurement that discriminates between the competing accounts.

Keywords: incretin; GLP-1 receptor agonist; tirzepatide; semaglutide; type 2 diabetes; obesity; energy balance; glycosuria; vagal afferent; autonomic neuropathy

Key points

- Across eight matched comparisons of five incretin-based molecules (same dose, duration and estimand), placebo-adjusted weight loss in type 2 diabetes was a median 0.62 times (range 0.50–0.78) that in people without diabetes: an absolute deficit of 2–7 percentage points.
- The deficit is graded across normoglycaemia, prediabetes and diabetes, appears with GLP-1, GIP/GLP-1, amylin/GLP-1 and GIP-antagonist/GLP-1 agents, and lies in the drug-attributable component, not the placebo response.
- Appetite suppression by incretin agonists is preserved in type 2 diabetes, but its magnitude has never been compared with that in people without diabetes.
- Correcting hyperglycaemia returns urinary glucose energy and lowers resting energy expenditure; assumption-driven estimates indicate these drug-arm-specific offsets could account for about 1–6 kg by 68–72 weeks, but neither has been measured in an incretin trial.
- Impaired GIP signalling or intrinsic insulin-resistance-driven pharmacological resistance are the least supported explanations; a matched energy-balance study is the key missing experiment.

1. Introduction

Medicines that act on the glucagon-like peptide-1 (GLP-1) receptor were first developed to lower glucose. They amplify glucose-dependent insulin secretion and suppress glucagon. They later proved to be the most effective drug treatments for obesity, and newer molecules add activity at the glucose-dependent insulinotropic polypeptide (GIP), glucagon or amylin receptors[1,2]. The same drugs are therefore used for two related but distinct goals: correcting dysglycaemia and reducing adiposity.

A consistent observation accompanies this dual use. When one molecule is studied at the same dose and for the same duration in separate trials of people with and without T2D, the trials in T2D report smaller weight reductions. For semaglutide 2.4 mg the placebo-adjusted reduction at 68 weeks was 12.4 percentage points without diabetes[3] and 6.2 points with T2D[4]. Similar differences appear with tirzepatide[5,6], with the oral small-molecule agonist orforglipron[7,8] and with cagrilintide–semaglutide[9,10].

The difference matters clinically, because weight reduction of this size changes glycaemic, cardiovascular and functional outcomes in T2D, and smaller responses shape both expectations and choice of agent. It is also informative mechanistically. Glucose lowering and weight reduction are separate pharmacological outputs. Correcting hyperglycaemia depends mainly on islet effects, whereas weight reduction depends on a sustained negative energy balance. The mechanisms that correct dysglycaemia need not determine weight loss, and correcting dysglycaemia can itself change energy balance.

This review addresses two questions separately. The clinical question is how confidently we can conclude that T2D is associated with less weight loss on comparable incretin-based therapy, and how large the difference is. The mechanistic question is which physiological or pharmacological processes could explain the difference, and which have actually been shown to do so. We consider four classes of explanation: (1) direct modification by T2D of the response to receptor agonism, whether centrally or through the peripheral secretory, degradative and afferent machinery that surrounds the receptor; (2) indirect metabolic effects of glycaemic correction on energy balance; (3) clinical and behavioural factors, including co-prescribed medicines; and (4) methodological artefacts arising from comparisons between trials. Throughout, we distinguish established causal links, supportive human evidence, indirect inference and hypotheses.

The evidence was assembled through structured searches of PubMed/MEDLINE and ClinicalTrials.gov, with backward citation tracking from pivotal trials. Every quantitative claim was checked against the primary abstract or, where accessible, the full text. This is a narrative review; it is not a systematic review and makes no claim of exhaustive screening (see Supplementary Methods and the quality-control report).

2. Clinical evidence: is weight loss attenuated in T2D?

2.1 What kind of comparison can answer the question?

Three kinds of evidence bear on the question, and they differ in strength. The strongest would be a randomised trial that stratifies by diabetes status and formally tests whether the treatment effect differs between strata (effect modification). No such trial has been designed for weight as its primary question. The next best are comparisons in which one protocol, or harmonised protocols, enrolled both populations, ideally analysed together from individual participant data (IPD). Weakest, but most plentiful, are comparisons of separate trials of the same molecule, dose, duration and estimand within one development programme. Comparisons of mean weight change between unrelated trials cannot distinguish an effect of diabetes from differences in population, background therapy or trial conduct. They also cannot show that individual patients respond differently, as opposed to populations differing on average.

2.2 Matched programmes

Table 1 summarises the matched comparisons, and Figure 1A shows them graphically. All use treatment-policy or treatment-regimen estimands, which count outcomes regardless of discontinuation, so the comparisons are not distorted by differences in persistence.

For semaglutide 2.4 mg weekly over 68 weeks, STEP 1 (no diabetes; baseline weight 105.3 kg, BMI 37.9 kg/m², HbA1c 5.7%) reported −14.9% versus −2.4% with placebo, an estimated treatment difference (ETD) of −12.4 percentage points[3]. STEP 2 (T2D; HbA1c 7–10%) reported −9.6% versus −3.4%, an ETD of −6.2 points[4]. The same pattern recurs at a higher dose in a later pair of trials from the same programme. STEP UP (no diabetes, n=1407; baseline weight 113.0 kg, BMI 39.9 kg/m², 73.7% women) reported −18.7% with semaglutide 7.2 mg versus −3.9% with placebo, an ETD of −14.8 points[11]. STEP UP T2D (n=512; baseline weight 110.1 kg, BMI 38.6 kg/m², HbA1c 8.1%, 51.8% women) reported −13.2% versus −3.9%, an ETD of −9.3 points[12]. This is the cleanest matched pair available: same sponsor, dose, duration and estimand, and an *identical* placebo-arm weight change of −3.9% in both populations. For tirzepatide over 72 weeks, SURMOUNT-1 (no diabetes) reported −19.5% at 10 mg and −20.9% at 15 mg versus −3.1% with placebo[5]. SURMOUNT-2 (T2D; mean HbA1c 8.02%, diabetes duration 8.5 years) reported −12.8% and −14.7% versus −3.2%[6]. For orforglipron 36 mg over 72 weeks the results were −11.2% versus −2.1% without diabetes[7] and −9.6% versus −2.5% with T2D[8]. For cagrilintide–semaglutide over 68 weeks they were −20.4% versus −3.0% (ETD −17.3)[9] and −13.7% versus −3.4% (ETD −10.4)[10]. Liraglutide 3.0 mg showed the same direction: a placebo-adjusted −5.6 kg without diabetes[13] and −4.2 kg with T2D[14], from similar baseline weights (about 106 kg).

Several other programmes point the same way but cannot be matched as cleanly. With retatrutide, a GIP/GLP-1/glucagon receptor tri-agonist, phase 2 weight loss reached 24.2% at 48 weeks in obesity[15] and 16.9% at 36 weeks in T2D[16]. The durations differ, so the comparison is only qualitative. Oral semaglutide 50 mg produced a 15.1% reduction (ETD −12.7) in people without diabetes[17]. The corresponding T2D trial of 25 and 50 mg had no placebo arm[18]. In the SURPASS and SUSTAIN programmes in T2D, doses were chosen for glycaemic control and comparators were active. Weight reductions were 7.0–9.5 kg with tirzepatide monotherapy over 40 weeks[19], 1.9–5.5 kg greater with tirzepatide than with semaglutide 1 mg[20], and only 0.9 kg greater with semaglutide 2.0 mg than with 1.0 mg[21]. These T2D trials show that weight response rises with dose and potency in diabetes. They do not provide a matched comparator without diabetes.

Across the nine percentage-based comparisons, the placebo-adjusted reduction in T2D was a median 0.63 times that without diabetes (range 0.50–0.78). The absolute deficit was a median 5.5 percentage points (range 1.8–6.9) (Table 1; analysis/derived.py). Two features stand out. First, the ratio is broadly similar for selective GLP-1 receptor agonists, the GIP/GLP-1 co-agonist tirzepatide and the amylin–GLP-1 combination. Second, placebo-arm weight change was not smaller in T2D (−3.4% vs −2.4% for semaglutide 2.4 mg; −3.9% vs −3.9% for semaglutide 7.2 mg; −3.2% vs −3.1% for tirzepatide; −2.5% vs −2.1% for orforglipron; −3.4% vs −3.0% for cagrilintide–semaglutide). The deficit therefore lies in the drug-attributable component rather than in the response to the lifestyle programme delivered to both arms.

2.3 Harmonised and single-protocol evidence

The STEP-HFpEF programme provides the closest available approach to a direct test. Two trials with harmonised eligibility criteria, endpoints and analyses enrolled people with obesity-related heart failure with preserved ejection fraction, one without diabetes and one with T2D. Both gave semaglutide 2.4 mg for 52 weeks. Weight change was −13.3% versus −2.6% without diabetes (ETD −10.7)[22] and −9.8% versus −3.4% with T2D (ETD −6.4)[23]. The prespecified IPD pooled analysis found substantial heterogeneity between the trials for weight ($p<0.0001$; I^2 95.8%), whereas improvements in heart-failure symptoms were consistent[24]. The authors note roughly 40% less weight loss with diabetes. Even here, diabetes status was not randomised, and the trials differed in recruitment and background therapy.

The phase 2 trial of maridebart cafraglutide, which combines GLP-1 receptor agonism with GIP receptor antagonism, enrolled an obesity cohort and an obesity-with-T2D cohort under one protocol[25]. At 52 weeks, weight change ranged from −12.3% to −16.2% (placebo −2.5%) in the obesity cohort and from −8.4% to −12.3% (placebo −1.7%) in the T2D cohort. The dosing arms were not identical between cohorts, and the T2D cohort was older (55.1 vs 47.9 years) and included fewer women (42% vs 63%).

Observational data point the same way. In a referral-centre cohort treated with semaglutide 1.7–2.4 mg, people with T2D lost 7.2% versus 11.8% at 6 months[26]. In a large propensity-matched electronic-record cohort, people without T2D lost more weight than people with T2D on both tirzepatide and semaglutide. The authors explicitly raise differences in motivation and concurrent weight-loss activity as possible explanations[27].

2.4 A gradient with dysglycaemia

The gradient also appears *within* T2D, indexed by glycaemic control rather than by diabetes status. In SURMOUNT-2, weight reduction with tirzepatide varied meaningfully by baseline HbA1c category[28], and pooled patient-level analyses of the SUSTAIN programme found baseline characteristics including glycaemia to modify the weight response to semaglutide[29]. The direction is consistent: higher baseline HbA1c, smaller weight reduction. This within-disease gradient is the single most informative feature of the phenotype, because it is the common prediction of every mechanism that scales with glycaemia — the energetic offsets of Section 9 above all — and it is not predicted by an explanation resting on diabetes status as a category.

In a post hoc analysis of STEP 1, participants with prediabetes lost 13.7% with semaglutide (ETD −11.3)[30], compared with 14.9% in the whole semaglutide arm[3]. Assuming the overall mean is the weighted average of the subgroup means, normoglycaemic participants lost about 15.9%. Normoglycaemia (~15.9%), prediabetes (13.7%) and T2D (9.6%) therefore form a graded series on the same dose and duration (Figure 1B). The gradient is unadjusted and comes from different trials, so it does not establish dose–response causation. It does argue against a threshold effect confined to overt diabetes. Its presence in prediabetes, where glycosuria is negligible, also limits how much of the gap any glycosuria-based explanation alone can carry (Section 9). In SURMOUNT-1, participants with prediabetes still lost 18.7–19.7% with tirzepatide 10–15 mg over 176 weeks[31], so large responses remain achievable in dysglycaemia.

2.5 The gap is not specific to incretin drugs

A phenotype confined to incretin pharmacology would point to the receptor. A phenotype that appears whenever weight is lost by any means points to the host. The evidence favours the second, and this bears directly on which mechanisms in Sections 5 to 10 can be primary.

Smaller weight reductions in T2D have been reported with three non-incretin pharmacotherapies acting on entirely unrelated targets. Naltrexone–bupropion produced −5.0% versus −1.8% with placebo in T2D (ETD −3.2)[32], against larger reductions in the non-diabetes COR programme[33]. Lorcaserin, a 5-HT2C agonist, produced −4.5% to −5.0% versus −1.5% in BLOOM-DM (ETD about −3.0)[34], again below the non-diabetes BLOOM result[35]. Phentermine–topiramate shows the same direction[36]. Bariatric surgery likewise: preoperative diabetes is a recognised predictor of smaller long-term weight loss[37]. None of these acts on the GLP-1 or GIP receptor, and surgery is not a drug at all.

Dietary restriction is the informative exception, and it cuts both ways. In a 36-day metabolic-ward study at a fixed 300 kcal/day, participants with T2D lost 308 ± 19 g/day versus 450 ± 31 g/day in non-diabetic participants ($p<0.01$) — but the non-diabetic group began with significantly greater body weight and lean mass, which the authors note as a confounder[38]. Behavioural programmes showed the same direction[39]. Against this, a systematic review and meta-analysis of five low-energy formula-diet studies (n=569) found *no* significant difference in final weight loss between people with and without T2D: −1.2 kg (95% CI −4.1 to 1.6), with near-identical rates of loss (0.6 vs 0.5 kg/week)[40].

Two readings of that null are possible. It may mean the gap is specific to pharmacotherapy and surgery and absent under enforced energy restriction. Or the meta-analysis may be underpowered and dominated by individual studies, as has been argued[41]. We cannot settle this from the published data.

What generality does and does not license. The inference this evidence supports is narrower than it first appears, and it is worth stating precisely because it is easy to overdraw. That the gap appears across drug classes sharing no receptor, and after surgery, makes a defect *specific to incretin receptor pharmacology* an unlikely primary cause. It does not identify which mechanism is responsible. It licenses only the conclusion that the operative mechanism lies in the host rather than in the drug — and at least three host-level classes of mechanism are compatible with everything above:

1. **Energetics of glycaemic correction** (Section 9). Any intervention that lowers weight in T2D also corrects glycaemia, so the glycosuric and resting-expenditure offsets apply regardless of how weight loss was achieved.
2. **A generalised deficit in anorectic responsiveness** (Section 4.3). Naltrexone–bupropion, lorcaserin and phentermine–topiramate all act centrally on appetite. If the diabetic state blunts the brain's response to satiation signals in general, every one of them underperforms, and so does any incretin agent.
3. **Degraded shared downstream machinery** (Section 5). All of these interventions, bariatric surgery included, ultimately route through common gut–brain afferent and hindbrain–hypothalamic pathways. Damage to that shared path blunts whatever is fed into it, whichever receptor opens the door.

These three are not mutually exclusive, and cross-intervention generality cannot distinguish among them, because all three predict it. Section 11.1 sets out the observations that would. Note also that bariatric surgery, which we cited above as evidence against a receptor-level cause, works substantially through gut-hormone remodelling — large postoperative rises in GLP-1 and PYY. If T2D has degraded the apparatus that *receives* those signals, surgery underperforms for reasons that have nothing to do with energetics. The surgical finding is therefore neutral between hypotheses 1 and 3, not supportive of 1.

The Henry study is the one piece of evidence here that does discriminate, and it points toward energetics: intake was fixed by protocol, so a difference in appetite could not express itself, yet the diabetic participants still lost less — while their fasting glucose fell from 291 to 95 mg/dL, closing a large glycosuric leak (Section 9). Its confounder (greater baseline weight and lean mass in the non-diabetic group) and the opposing meta-analytic null[40] prevent it from settling the matter, but it is the right *kind* of experiment[42].

2.6 Could the difference be an artefact?

Several features of the trials argue against a purely methodological explanation.

- **Dose, duration and estimand** were matched within each programme.
- **Discontinuation and tolerability** do not favour the trials without diabetes. Gastrointestinal adverse events were reported by 72.5% in REDEFINE 2 versus 79.6% in REDEFINE 1[9,10]. Discontinuations for adverse events were similar in ATTAIN-1 and ATTAIN-2 (5.3–10.3% vs 6.1–9.9%)[7,8].
- **Analyses that remove discontinuation** leave the difference in place. Under the efficacy estimand, which estimates the effect had treatment continued as intended, SURMOUNT-2 reported ETDs of −10.1 and −12.4 points[6]. These remain well below the treatment-regimen ETDs of −16.4 and −17.8 points in SURMOUNT-1[5], although the treatment-regimen estimand includes the effect of discontinuation and so understates the drug's full on-treatment effect.
- **Drug exposure** is unlikely to explain the difference. Within T2D programmes, population pharmacokinetic analyses identify body weight as the only clinically relevant covariate of semaglutide exposure[43], and weight and HbA1c responses track plasma exposure[44]. Baseline body weights in the matched programmes were similar. An exposure–response model of semaglutide spanning populations with and without T2D used baseline HbA1c as a covariate[45], but we could not verify from the accessible text how much it changed the predicted response.

Other differences do favour the trials without diabetes. The T2D trials enrol older participants with a higher proportion of men. Women lost more weight in the STEP-HFpEF pool[24]; STEP 1 enrolled 74.1% women and SURMOUNT-2 50.7%[3,6]. T2D participants also continued glucose-lowering drugs, including sulfonylureas in 26.7% of SURMOUNT-2 participants[6]. No analysis has pooled IPD across programmes with adjustment for age, sex, baseline BMI, HbA1c, diabetes duration and background therapy. Without one, the share of the gap attributable to these population differences cannot be quantified.

2.7 What the clinical evidence establishes

The clinical phenotype is robust. It appears in every matched programme, in harmonised IPD, in a single-protocol trial and in observational cohorts, and it is graded with glycaemic status. On average, people with T2D achieve about 50–80% (median ~60%) of the placebo-adjusted weight reduction seen without diabetes. The absolute shortfall is about 2–7 percentage points at 52–72 weeks, larger with the most potent agents. Because the deficit occurs in the drug-attributable component, and across GLP-1 receptor agonists, GIP receptor agonism and antagonism, amylin co-agonism and lifestyle intervention[39], it is more likely to reflect host physiology and context than a defect specific to one receptor. What the evidence cannot yet establish is how much of the population difference is effect modification by diabetes itself rather than confounding by co-varying traits.

Table 1. Matched comparisons of incretin-based therapies in populations with and without T2D. Nine percentage-based comparisons; median ratio 0.63, range 0.50–0.78. Values are mean % change in body weight from baseline to end of treatment (drug vs placebo). ETD = estimated treatment difference (percentage points). TP = treatment-policy estimand; TR = treatment-regimen estimand.

Agent and dose	Weeks	Population without diabetes (trial; key baseline)	Drug vs placebo; ETD	Population with T2D (trial; key baseline; background therapy)	Drug vs placebo; ETD	Ratio T2D/no-T2D (ETD)
Semaglutide 2.4 mg weekly	68	STEP 1, n=1961; 105.3 kg, BMI 37.9, HbA1c 5.7%, 74% women[3]	−14.9 vs −2.4; −12.4 (−13.4 to −11.5), TP	STEP 2, n=1210; HbA1c 7–10%; 0–3 oral agents[4]	−9.6 vs −3.4; −6.2 (−7.3 to −5.2), TP	0.50
Semaglutide 7.2 mg weekly	72	STEP UP, n=1407; 113.0 kg, BMI 39.9, 73.7% women, age 47[11]	−18.7 vs −3.9; −14.8 (−16.2 to −13.4), TP	STEP UP T2D, n=512; 110.1 kg, BMI 38.6, HbA1c 8.1%, 51.8% women, age 56; ~30% on SGLT2i[12]	−13.2 vs −3.9; −9.3 (−11.0 to −7.7), TP	0.63
Semaglutide 2.4 mg weekly (HFpEF)	52	STEP-HFpEF, n=529[22]	−13.3 vs −2.6; −10.7 (−11.9 to −9.4)	STEP-HFpEF DM, n=616; HbA1c ≤10%[23]	−9.8 vs −3.4; −6.4 (−7.6 to −5.2)	0.60 (IPD heterogeneity p<0.0001)[24]
Tirzepatide 10 mg weekly	72	SURMOUNT-1, n=2539; 104.8 kg, BMI 38.0[5]	−19.5 vs −3.1; −16.4*, TR	SURMOUNT-2, n=938; 100.7 kg, BMI 36.1, HbA1c 8.02%, duration 8.5 y; metformin 89%, SU 27%, SGLT2i 20%[6]	−12.8 vs −3.2; −9.6 (−11.1 to −8.1), TR	0.59

Agent and dose	Weeks	Population without diabetes (trial; key baseline)	Drug vs placebo; ETD	Population with T2D (trial; key baseline; background therapy)	Drug vs placebo; ETD	Ratio T2D/no-T2D (ETD)
Tirzepatide 15 mg weekly	72	SURMOUNT-1[5]	−20.9 vs −3.1; −17.8*, TR	SURMOUNT-2[6]	−14.7 vs −3.2; −11.6 (−13.0 to −10.1), TR	0.65
Orforglipron 36 mg daily (oral)	72	ATTAIN-1, n=3127[7]	−11.2 vs −2.1; −9.1*, TR	ATTAIN-2, n=1613; 101.4 kg, BMI 35.6, HbA1c 8.05%[8]	−9.6 vs −2.5; −7.1 (−8.2 to −6.1), TR	0.78
Orforglipron 12 mg daily (oral)	72	ATTAIN-1[7]	−8.4 vs −2.1; −6.3*, TR	ATTAIN-2[8]	−7.0 vs −2.5; −4.5, TR	0.71
Cagrilintide 2.4 mg + semaglutide 2.4 mg weekly	68	REDEFINE 1, n=3417[9]	−20.4 vs −3.0; −17.3 (−18.1 to −16.6), TP	REDEFINE 2, n=1206; HbA1c 7–10%[10]	−13.7 vs −3.4; −10.4 (−11.2 to −9.5), TP	0.60
Maridebart cafraglutide (highest-response arm)	52	Obesity cohort, n=465; BMI 37.9, 63% women[25]	−16.2 vs −2.5; −13.7*, TP	Obesity–T2D cohort, n=127; BMI 36.5, 42% women[25]	−12.3 vs −1.7; −10.6*, TP	0.77 (dose arms differ)
Liraglutide 3.0 mg daily	56	SCALE Obesity & Prediabetes, n=3731; 106.2 kg[13]	−8.4 vs −2.8 kg; −5.6 kg	SCALE Diabetes, n=846; 0–3 oral agents[14]	−6.0% (−6.4 kg) vs −2.0% (−2.2 kg); −4.0 (−5.1 to −2.9)	0.75 (kg basis)

*ETD calculated as drug minus placebo from published means; confidence interval not reported in the abstract. SU, sulfonylurea; SGLT2i, sodium–glucose co-transporter 2 inhibitor. Ratios computed in analysis/derived.py.

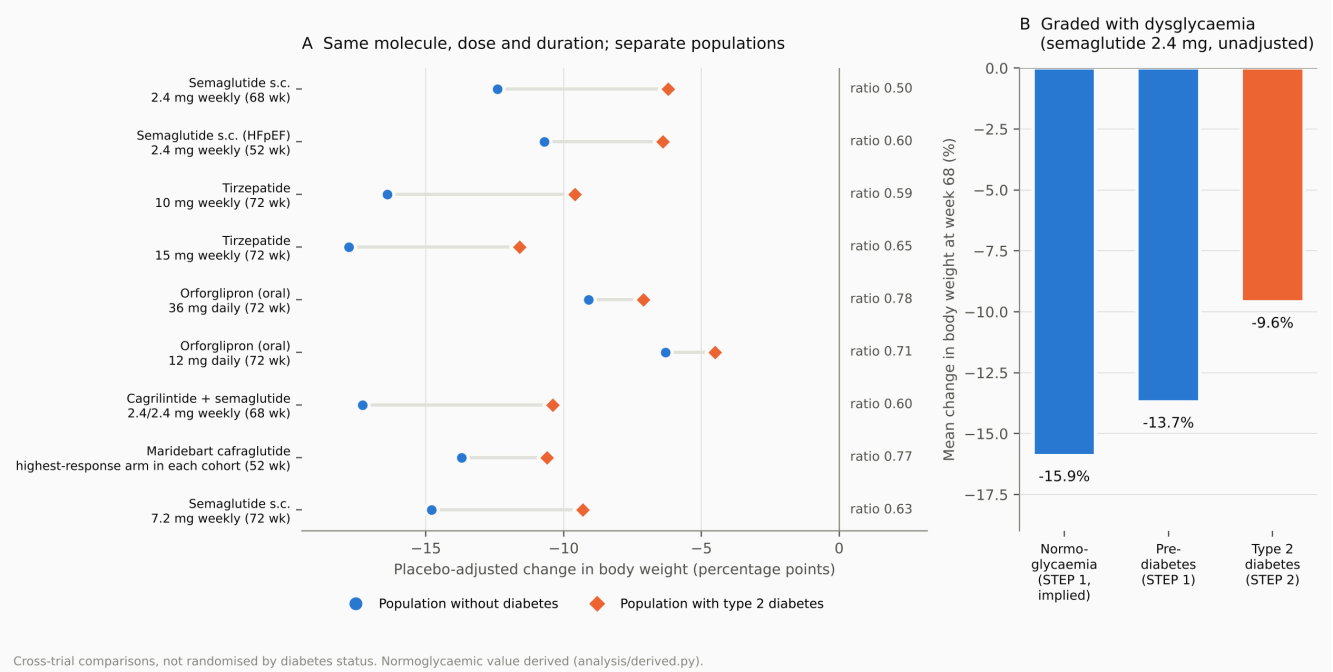


Fig. 1 | The clinical phenotype. (A) Placebo-adjusted change in body weight (estimated treatment difference, percentage points) for the same molecule, dose and duration in separate populations without diabetes (circles) and with T2D (diamonds). Values on the right are the ratio of the T2D to the non-diabetes effect. Sources are in Table 1. Comparisons are between trials and not randomised by diabetes status; the maridebart cafraglutide cohorts used different dose arms. (B) Mean weight change with semaglutide 2.4 mg at 68 weeks: normoglycaemia (implied from STEP 1 overall and prediabetes data; derived), prediabetes (STEP 1 post hoc subgroup) and T2D (STEP 2). Unadjusted cross-trial gradient. Source code: figures/make_figures.py.

3. Incretin pharmacology: separating glucose lowering from weight loss

3.1 Endogenous physiology versus pharmacology

Endogenous GLP-1 and GIP are short-lived peptides secreted after meals. Both potentiate glucose-stimulated insulin secretion. GLP-1 also suppresses glucagon, slows gastric emptying and reduces food intake[1]. In T2D the incretin effect, the extra insulin secreted after oral compared with intravenous glucose at matched glycaemia, is markedly reduced: incretin factors accounted for 36% of the insulin response in T2D versus 73% in matched controls[46]. The defect is largely a consequence of the diabetic state rather than its cause. It appears in diabetes secondary to chronic pancreatitis but not in pancreatitis with normal glucose tolerance[47]. Four weeks of near-normoglycaemia improved the late-phase β-cell response to GLP-1 and GIP three- to fourfold[48].

These findings concern the endocrine pancreas at physiological hormone concentrations. They do not imply reduced responsiveness to long-acting agonists at pharmacological exposures. Infused GLP-1 retained most of its insulinotropic activity in mild T2D, whereas the insulinotropic effect of GIP was reduced by 54%[49]. GIP also failed to amplify late-phase secretion[50]. Graded GLP-1 infusions increased β-cell glucose responsiveness in T2D, but at each dose the response was 3–5 times smaller than in controls. Even so, a low dose restored responsiveness to the level of untreated controls[51]. A reduced physiological incretin effect is therefore a statement about the islet. It says nothing direct about the brain and gut circuits through which agonists reduce energy intake.

3.2 Two outputs, partly separate pathways

The glucose-lowering effects of incretin drugs arise mainly from islet mechanisms (insulin secretion and glucagon suppression), from slower gastric emptying of meal glucose, and secondarily from weight reduction and improved insulin sensitivity. In a mechanistic study in T2D, tirzepatide improved the clamp disposition index, total insulin secretion, insulin sensitivity (M value) and glucagon suppression more than semaglutide 1 mg[52]. Weight reduction, by contrast, depends on a sustained negative energy balance. That balance is driven mainly by reduced energy intake, with little or no increase in energy expenditure in humans (Sections 4 and 10). Both outputs share the receptor but not the effector organs. This is why a drug can improve glycaemia without changing weight, as low-dose liraglutide (0.6 mg) did in an 8-week respiratory-chamber study in T2D[53], and why glycaemic correction can feed back on energy balance (Section 9).

3.3 Exposure, receptor engagement and molecular class

Long-acting peptide agonists achieve sustained plasma concentrations far above endogenous GLP-1. Within T2D, semaglutide exposure varies mainly with body weight[43]. Oral and subcutaneous formulations show nearly identical exposure–response relationships for HbA1c and body weight, which suggests that variation between individuals in GLP-1 receptor responsiveness is not a major determinant of response[44]. Agents differ in other ways. Tirzepatide engages the GIP receptor more than the GLP-1 receptor at clinical doses, and at the GLP-1 receptor it favours cAMP signalling over β-arrestin recruitment and causes less receptor internalisation[54]. Orforglipron is a non-peptide oral agonist[7]. Cagrilintide adds amylin receptor agonism[9]. Maridebart cafraglutide pairs GLP-1 agonism with GIP receptor antagonism[25]. Despite these differences, the attenuation ratio in T2D is similar across agents (Table 1). That similarity is the first indication that the explanation lies mainly in the host rather than in a receptor-specific pharmacodynamic defect.

4. Central regulation of appetite and energy intake

4.1 How incretin agonists reduce intake

Reduced energy intake is the dominant driver of incretin-induced weight loss. In adults with obesity, semaglutide 1.0 mg reduced ad libitum energy intake across a test day by 24%, with less hunger and food craving, better control of eating and a lower preference for high-fat foods. Resting metabolic rate adjusted for lean mass did not change[55]. Semaglutide 2.4 mg reduced ad libitum lunch intake by 35% after 20 weeks[56]. Tirzepatide reduced intake at an ad libitum lunch by about 525 kcal at week 3 in adults without diabetes. It also reduced food cravings, the tendency to overeat and responsiveness to food cues, and reduced activation to high-fat, high-sugar food images in orbitofrontal, cingulate and hippocampal regions[57]. In humans, GLP-1 receptors have been identified on neurons of the hypothalamus, medulla and parietal cortex[58].

Animal studies define the circuitry. Peripherally administered semaglutide does not cross the blood–brain barrier. It reaches circumventricular organs and periventricular sites, activates hindbrain and hypothalamic neurons, and engages secondary nodes such as the lateral parabrachial nucleus. In rodents it reduces intake without lowering energy expenditure[59]. Liraglutide directly activates arcuate POMC/CART neurons and indirectly inhibits NPY/AgRP neurons through GABAergic transmission. Its weight-lowering effect persisted after loss of GLP-1 receptors in vagal afferents, area postrema or paraventricular nucleus[60]. Other work shows that vagal afferent and central receptor populations both contribute, because vagal deafferentation shifts the dose–response curve[61]. The pharmacological effect is also separable from the endogenous brainstem preproglucagon system. Those neurons are the predominant source of GLP-1 within the brain, but their ablation left ad libitum intake, body weight and glucose tolerance unchanged in mice, affecting intake only after a prolonged fast or a liquid preload and abolishing stress-induced hypophagia — a secondary satiation circuit rather than a primary regulator[62] — and they are not required for the anorexia induced by GLP-1 receptor agonists[63]. These rodent data describe pathways. They do not show that any pathway behaves differently in human T2D. The peripheral limb of the same circuitry, including vagal afferent transmission and the endogenous incretin background on which the drug is superimposed, is considered separately in Section 5.

4.2 Appetite suppression in T2D: present, magnitude untested

Three types of evidence should be kept apart.

Incretin agonism suppresses appetite in T2D. This is well supported. Intravenous GLP-1 reduced energy intake by 27% in T2D[64]. Liraglutide 1.8 mg reduced ad libitum intake by 18% after 3 weeks[65]. Tirzepatide and semaglutide both reduced appetite in T2D over 28 weeks[66]. In a crossover fMRI study, exenatide reduced food intake and food-cue responses in insula, amygdala, putamen and orbitofrontal cortex in obese people with T2D as well as in normoglycaemic obese people, and receptor blockade with exendin 9-39 largely prevented these effects[67]. In T2D, liraglutide reduced insula and putamen responses to food pictures after 10 days but not after 12 weeks, relative to insulin glargine[68]. It also reduced parietal activation to highly desirable foods[58].

Whether appetite suppression is smaller in T2D is unknown. No study has compared ad libitum intake, appetite ratings or food-cue responses between people with and without diabetes receiving the same agonist under identical conditions. Apparent differences between studies, such as 27% with GLP-1 infusion in T2D versus 12% in healthy men[69], reflect different infusion rates, meals and populations and cannot be interpreted as effect modification.

Whether a difference in appetite suppression explains the weight-loss gap has therefore never been tested. One observation is suggestive. In T2D, tirzepatide produced more weight loss than semaglutide 1 mg, but intake at an ad libitum lunch did not differ between the drugs. The authors concluded that measured intake did not account for the difference in weight outcome[66]. A single test meal is an insensitive measure of 24-hour intake, so this finding should not be overinterpreted.

4.3 Could the diabetic state blunt central responses?

This is one of the three host-level hypotheses that the cross-intervention evidence of Section 2.5 leaves open, and it deserves more weight than the absence of direct tests might suggest. Several lines of human evidence show that the diabetic brain responds abnormally to nutrient and food signals.

The most direct is a functional MRI study of nutrient sensing. Glucose ingestion produces a prolonged inhibition of hypothalamic neuronal activity in healthy people. In seven men with T2D compared with ten age-matched healthy men, that inhibition **failed to occur**[70]. The groups were matched for body mass index (27.9 vs 26.1 kg/m²), so this is a feature of the diabetic state rather than of adiposity. The authors concluded that failure of neural circuits to adapt properly to nutrient ingestion may contribute to metabolic imbalance in T2D. The study is small, men only, and from 2007, and it used oral glucose rather than a pharmacological agonist — but it is a direct demonstration in humans that central nutrient sensing is impaired in diabetes.

Converging markers point the same way. In adults with obesity, MRI markers of mediobasal hypothalamic gliosis were greater in impaired glucose tolerance and T2D than in normal glucose tolerance, independent of adiposity, and correlated with glycaemia and insulin resistance[71]. Brain insulin resistance in obesity and T2D alters eating behaviour and reward processing[72]. In obese T2D and normoglycaemic obese participants, food-cue responses in appetite and reward regions were greater than in lean participants[67], and postprandial neural processing of food cues varies with body weight and with circulating insulin[73]. A larger baseline drive to eat, or a hypothalamus that does not register ingested nutrient, would oppose a given pharmacological suppression.

Two features make this hypothesis worth taking seriously alongside the energetic account. First, it predicts the gap for *any* anorectic agent, which is what Section 2.5 observes. Second, and more tellingly, the abnormalities above are present in **impaired glucose tolerance as well as overt diabetes**[71]. That matters because the weight-loss deficit is already visible in prediabetes (Section 2.4), where glycosuria is negligible and the energetic account has least to offer. The one gradient the energetic hypothesis cannot explain is precisely the one where this hypothesis is strongest.

What is still missing is the direct test. No study has compared the *magnitude* of agonist-induced anorexia between people with and without diabetes under matched conditions, so whether gliosis, central insulin resistance, impaired hypothalamic nutrient sensing or heightened food-cue reactivity reduces the size of the drug effect remains untested. These are well-motivated hypotheses with human mechanistic support, not established mediators.

4.4 Compensatory regulation

Weight loss provokes compensatory increases in appetite. In patients treated with canagliflozin, whose energy deficit comes from glycosuria without appetite suppression, calculated energy intake rose by about 100 kcal/day for each kilogram lost. That is more than three times the accompanying adaptation in energy expenditure[74]. A common interpretation, not directly tested in humans, is that incretin agonists shift the level at which this feedback operates. Under that model, any additional, drug-independent energy inflow in T2D, such as glucose energy previously lost in urine (Section 9), would act on this feedback system and lead to a higher weight plateau.

5. Peripheral incretin signalling: the endogenous background, its degradation and afferent transmission

Section 4 considered the brain. This section considers everything upstream of it: the endogenous incretin secreted by the gut, the enzymes that destroy it, the receptors it meets in the gut wall and hepatoportal bed, and the afferent nerves that carry the resulting signal. Each of these is altered in T2D. Whether any of it changes how a pharmacological agonist performs is a separate question, and the answer turns on a single issue of pharmacology that we take first.

5.1 Why the endogenous background might matter at all

A long-acting agonist does not need endogenous GLP-1 to activate the GLP-1 receptor. It reaches sustained plasma concentrations far above physiological ones (Section 3.3), and it is engineered to resist the enzymatic degradation that limits the native peptide[75]. On the simplest reading, the state of the endogenous system is therefore irrelevant to drug effect, and reduced native secretion in T2D would predict a *larger* relative response to an agonist, not a smaller one.

Two considerations complicate that reading. First, endogenous GLP-1 acts substantially where it is released. Native GLP-1 has a circulating half-life of minutes, and a large fraction of an intravenous bolus is inactivated before it leaves the portal circulation[76,77]. Much of its physiological action is therefore paracrine and neural, exerted on receptors in the gut wall and hepatoportal bed before the peptide reaches the systemic circulation at all[76]. A drug delivered subcutaneously arrives at those same receptors from the systemic side, at different concentrations and without the postprandial timing. Second, the pathway that carries the signal onward — vagal afferent fibres and their brainstem targets — is itself a structure that diabetes damages (Section 5.5). If the drug's effect depends partly on that pathway, then degrading the pathway could degrade the drug effect regardless of receptor occupancy.

The peripheral hypothesis, then, is not that T2D leaves too little endogenous GLP-1 for the drug to work with. It is that T2D may erode a paracrine and neural substrate on which part of the agonist's anorectic effect depends, and may remove a physiological signal that normally works alongside it.

5.2 Endogenous GLP-1 and GIP secretion in T2D: contested, and graded with glycaemia

Individual studies support reduced GLP-1 secretion in T2D. In 54 patients with T2D compared with 33 controls with normal glucose tolerance, the 4-hour meal GLP-1 response was lower (area under the curve 2482 ± 145 versus 3101 ± 198 pmol/L × 240 min; p = 0.024), and a model including diabetes, BMI, male sex and other terms explained 42% of the variance in that response[78]. In a meal study of eight obese patients with T2D against eight matched obese healthy participants, GLP-1 responses were reduced in T2D while GIP responses were normal[79].

Systematic evidence qualifies this substantially. A meta-analysis of 22 trials and 29 stimulation tests (275 patients with T2D, 279 controls) found *no* significant difference in total GLP-1 responses by any measure — peak concentration, total or incremental area under the curve, or their time-corrected equivalents[80]. A companion meta-analysis of 23 trials found GIP secretion likewise preserved[81]. The authors concluded that patients with T2D do not in general show reduced incretin secretion.

The qualification that survives is the one that matters most here. In both meta-analyses, meta-regression showed that HbA1c and fasting plasma glucose predicted the incremental GLP-1 response, and HbA1c predicted GIP responses negatively[80,81]. Reduced secretion is thus not a property of having diabetes but a function of how poorly controlled it is. This matters for two reasons. It means a uniform secretory deficit cannot be assumed in trial populations, whose mean HbA1c was around 8% (Table 1). And it predicts a graded rather than threshold relationship, which is the shape the clinical phenotype actually has (Section 2.4). The same logic applies at the islet: near-normalisation of glycaemia for four weeks improved the β-cell response to both GLP-1 and GIP three- to fourfold[48], and in rat islets chronic hyperglycaemia reduced GLP-1 and GIP receptor expression reversibly[82].

5.3 Degradation: DPP-4 and neprilysin

Dipeptidyl peptidase-4 (DPP-4) cleaves GLP-1(7–36) amide to an inactive form and is expressed on endothelium throughout the portal and systemic circulation[76]. Higher DPP-4 activity would lower the active fraction of endogenous GLP-1 reaching paracrine and portal receptors — a second reduction compounding any secretory deficit.

The human evidence is weaker than it is often presented. In 985 people with newly diagnosed T2D, plasma DPP-4 activity correlated positively with HbA1c and HOMA-IR, and those in the highest activity quartile had worse metabolic profiles[83]. This is a cross-sectional association within T2D; it does not establish that activity is higher than in people without diabetes. A 12-week exercise and weight-loss intervention in 14 older adults with metabolic syndrome lowered plasma DPP-4 by 10%, and the fall correlated with improved clamp-derived insulin sensitivity ($r = -0.59$)[84] — an intervention study with no non-diabetic comparator, n = 14, and no measurement of GLP-1 or body weight. Neither study supports a quantitative claim about how much active GLP-1 is lost.

Two mechanistic observations are nonetheless informative about *where* degradation matters. In mice, blood-glucose lowering after the DPP-4 inhibitor anagliptin correlated with small-intestinal DPP-4 inhibition ($r = 0.949$) but not with plasma DPP-4 inhibition, in both normal and diabetic models, with intact GLP-1 rising in the portal circulation[85]. And neprilysin, upregulated in obesity and T2D, also degrades GLP-1: high-fat-fed neprilysin-deficient mice had higher active GLP-1 (5.8 ± 1.1 versus 3.5 ± 0.8 pmol/L), lower plasma DPP-4 activity and better glucose tolerance[86]. Both findings locate the degradative bottleneck in the gut and portal bed rather than the systemic circulation. Both are rodent studies, and the neprilysin result is a genetic deletion, not pharmacological inhibition.

The clinical counterpoint is decisive for the drug question. DPP-4 inhibitors raise endogenous active GLP-1 and improve glycaemia, yet produce negligible weight loss. Prolonging the residence of native GLP-1 is therefore not sufficient to reduce body weight, which bounds how much any degradation-based mechanism can contribute to a weight-loss deficit.

5.4 Where peripheral GLP-1 acts: paracrine vagal afferents, not the hepatoportal route

GLP-1 receptors are expressed on vagal afferent neurons of the nodose ganglion, which project to the nucleus tractus solitarii and onward to hypothalamic feeding centres[61,77]. Three lines of animal work define the anatomy more precisely, and they do not all point the same way.

A portal glucose sensor exists and is GLP-1-receptor-dependent. Low-rate intragastric glucose in mice increased muscle glycogen synthesis by 30% and induced c-Fos in brainstem and hypothalamus, colocalising with arcuate NPY neurons. The response was diminished by central exendin 9-39 and abolished in GLP-1 receptor knockout mice — and, importantly, high-fat feeding prevented it entirely[87]. Related work implicates intestinal gluconeogenesis in generating the portal glucose signal detected by periportal afferents[88]. Note that the receptor dependence demonstrated in this model was *central*, not portal.

But the hepatoportal branch is not the required route. Selective ablation of the common hepatic branch of the vagus did not abolish the GLP-1-receptor-mediated incretin response or GLP-1's suppression of intake in rats, whereas complete subdiaphragmatic vagal deafferentation did[89]. The authors concluded that what matters is paracrine signalling on GLP-1 receptors borne by vagal afferents of general gastrointestinal origin, not the hepatoportal bed specifically.

And circulating GLP-1 may not need vagal afferents at all. Intravenous GLP-1 reduced sucrose intake in rats undiminished by vagotomy or capsaicin treatment — in fact enhanced at the highest dose — with comparable hindbrain c-Fos induction in all groups[90]. The authors inferred that vagal participation is limited to paracrine effects near the sites of secretion, and that endocrine GLP-1 acts elsewhere. Conversely, the anorectic and glucose-tolerance effects of the GLP-1-releasing sweetener D-allulose were blunted by vagotomy, GLP-1 receptor blockade and vagal-afferent-selective GLP-1 receptor inactivation[91], consistent with a vagal route for *locally released* GLP-1.

The most important finding for the present argument concerns receptor identity. In anaesthetised rats, intraportal GLP-1(7–36) amide at both periphysiological (0.2 pmol) and pharmacological (4.0 pmol) doses increased hepatic vagal afferent discharge. Exendin-4 did not, at any dose up to 40 pmol, and exendin 9-39 at more than 100-fold molar excess failed to block the response to GLP-1(7–36) amide[92]. The hepatic vagal reception of portal GLP-1 therefore appears to operate through a receptor

pharmacologically distinct from the classical GLP-1 receptor — one that a GLP-1 receptor agonist does not engage. Consistent with this, intraportal exenatide in conscious dogs produced no insulin-independent effect on hepatic glucose metabolism[93].

This is a problem for the peripheral hypothesis, not a support for it, and it should be stated plainly. If the portal afferent pathway is not activated by exogenous agonists, then loss of endogenous portal signalling in T2D removes a physiological contribution that the drug was never supplying, in both populations. It would not blunt the drug's effect; it would lower the baseline on which the drug acts. The receptor in question has also never been molecularly identified, the experiments are from 2000 in small numbers of anaesthetised rats, and no one has repeated them with a modern long-acting agonist. The claim should be treated as unresolved rather than established.

5.5 Human evidence that intact vagal innervation matters

One human study bears directly on whether peripheral neural transmission is necessary for GLP-1's effect on eating. Twenty men with truncal vagotomy and pyloroplasty and ten matched healthy controls received GLP-1(7–36) amide or saline during a standardised meal followed by an ad libitum meal. GLP-1 significantly reduced ad libitum food intake in controls but had no effect in the vagotomised group. It delayed gastric emptying in controls but not after vagotomy, and left insulin secretion unaffected in the vagotomised group while suppressing it in controls. Postprandial glucose fell by the same percentage in both groups, and peak postprandial GLP-1 concentrations were roughly fivefold higher after vagotomy[94].

This is the strongest available human evidence that intact vagal innervation is required for exogenous GLP-1 to reduce food intake. Its limitations are material. The participants were men with surgical vagotomy and pyloroplasty for ulcer disease, a far more complete and anatomically different lesion than diabetic autonomic neuropathy; pyloroplasty itself accelerates gastric emptying and confounds the intake measurement; the infusion delivered native GLP-1, not a long-acting agonist engineered to act systemically; and the study was not designed to measure body weight over time. It establishes that the vagal route can be rate-limiting for native GLP-1 in humans. It does not establish that partial vagal damage limits the effect of semaglutide or tirzepatide.

5.6 Vagal and enteric neuropathy in diabetes

Autonomic neuropathy is common in T2D and begins early. In 806 patients with newly diagnosed T2D undergoing systematic assessment, cardiovascular autonomic neuropathy was present in 18.6% and somatic neuropathy in 21.2%[95]. Reported prevalence across populations and diagnostic methods ranges very widely, from a few per cent to the large majority, depending on the tests applied and the cohort studied[96]. Cardiovascular autonomic function is a proxy for vagal integrity, not a measure of the gastrointestinal afferent limb.

Direct evidence that the afferent limb is affected is largely preclinical or from type 1 diabetes. A post-mortem study found severe loss of myelinated fibres in the vagus nerve and sympathetic trunks, with ganglion cell changes and inflammatory infiltrates — but in five cases of early-onset *insulin-dependent* diabetes with established clinical autonomic failure[97], which limits its transfer to typical T2D trial populations. In db/db mice, morphometric fibre loss and shrinkage appeared across peripheral nerves, though changes in the vagus appeared late relative to other nerves[98].

Two animal studies address function in type 2 models specifically, and they are the most relevant of this group. In UCD-T2DM rats, orogastric glucose activated duodenal enterochromaffin, K and L cells, enteric neurons and vagal afferent neurons in prediabetic animals, but this activation was markedly reduced in recent-onset and 3-month diabetic animals, with reduced GLP-1 immunoreactivity (and preserved GIP)[99]. In insulin-resistant obese animals studied with a gallium-labelled GLP-1 receptor PET probe and single-neuron electrophysiology, portal vagal afferents failed to modulate firing during glucose infusion, alongside reduced GLP-1 receptor binding potential, most marked between the splenic vein and the hepatic inlet[100]. Both show that the gut–vagal glucose-sensing pathway degrades with the diabetic and insulin-resistant state. Neither administered a GLP-1 receptor agonist or measured body weight.

5.7 The one study that tests the drug question — and what it does and does not show

Only one study we identified blunted an *agonist's* effect alongside peripheral pathway changes. In obese-prone and obese-resistant rats, intraperitoneal exendin-4 suppressed intake effectively in both strains on chow. On a high-energy, high-fat diet, obese-prone rats suppressed intake significantly less than obese-resistant rats at every dose tested. This was accompanied by downregulated GLP-1 receptor mRNA in the vagal nodose ganglia, lower plasma GLP-1, reduced intestinal GLP-1 protein and a reduced number of L cells in the distal ileum[101].

This is the closest existing analogue of the clinical observation: a dysmetabolic host showing reduced anorectic response to a GLP-1 receptor agonist, with a peripheral mechanism measured in the same animals. It is also a diet-induced obesity model rather than a diabetes model, the agonist is short-acting and given intraperitoneally, the endpoint is acute intake rather than sustained weight change, and the design cannot separate nodose receptor downregulation from the several other changes present.

5.8 What the peripheral axis can and cannot carry

Taken together, the peripheral evidence supports a coherent hypothesis and falls well short of establishing it.

In favour: the gut–vagal afferent pathway is measurably degraded in type 2 diabetic and insulin-resistant animals[99,100]; intact vagal innervation is required for native GLP-1 to reduce intake in humans[94]; agonist-induced anorexia is blunted in a dysmetabolic rodent host with nodose GLP-1 receptor downregulation[101]; and secretory and islet responses recover when glycaemia is corrected[48,80], which fits the graded clinical phenotype.

Against, or limiting: incretin secretion is on average preserved in T2D, with reduction confined to poorer glycaemic control[80,81]; the portal afferent receptor appears not to be engaged by agonists at all[92,93]; the hepatoportal branch is not the required route even for native GLP-1[89]; circulating GLP-1 can suppress intake without vagal afferents[90]; central GLP-1 receptor populations are sufficient for agonist-induced anorexia in rodents[59,60]; and prolonging native GLP-1 with DPP-4 inhibition does not produce weight loss.

Three further arguments bound the contribution — but one apparent bound is not one, and it is worth separating them carefully.

The cross-intervention evidence does not bound this hypothesis; it depends on which version is meant. The gap appears with naltrexone–bupropion, lorcaserin and phentermine–topiramate, none of which touches the GLP-1 or GIP receptor, and after bariatric surgery (Section 2.5)[32,34,36,37]. This does rule out the *narrow* version of the peripheral hypothesis — that reduced endogenous GLP-1, raised DPP-4 activity or lost portal GLP-1 signalling is the primary cause — because none of those is engaged when no incretin drug is given.

It does not touch the *broad* version, and arguably supports it. The vagal afferent pathway described in Sections 5.4 to 5.6 is not an incretin-specific channel. The same fibres carry cholecystokinin, PYY, ghrelin, distension and nutrient signals, and the same hindbrain and hypothalamic nodes integrate them[77,102]. A degraded afferent path blunts whatever is fed into it. That predicts a deficit with every anorectic agent, and with bariatric surgery — which works substantially through large postoperative rises in GLP-1 and PYY, and therefore depends on an intact receiving apparatus. The generality of the phenotype is thus as consistent with damage to shared gut–brain machinery as with the energetics of glycaemic correction. Our earlier framing of this evidence as a bound on the present section was mistaken; it bounds only the incretin-specific reading.

The honest state of this literature, however, is that the multi-axis version has not been tested either. A comprehensive review of ghrelin, CCK, GLP-1 and PYY concludes that while CCK is established as an endogenous endocrine control of eating in healthy-weight people, the roles of all four hormones in eating in obesity and after bypass surgery remain uncertain, with local rather than endocrine signalling likely and no good method for assessing it[102]. One T2D-specific study illustrates how interconnected these axes are: in metformin-treated patients with T2D, cholecystokinin-induced gallbladder emptying stimulated GLP-1 secretion through bile acid release, and a single dose of the bile acid sequestrant sevelamer abolished that stimulation[103]. Bile acid, CCK and L-cell signalling are one system, not three, and diabetes perturbs all of it.

Of the genuine bounds, the first is that the cross-agent test used in Section 7.3 applies here too. If the deficit depended on the endogenous GLP-1 background or its degradation, it should be smaller for agents that bypass both. Orforglipron is a non-peptide small molecule that is not a DPP-4 substrate, and cagrilintide–semaglutide adds amylin receptor agonism; both show the deficit (ratios 0.71–0.78 and 0.60; Table 1). The attenuation is not confined to agents whose action depends on the native incretin axis. Second, almost all participants in the T2D trials were taking metformin — 89% in SURMOUNT-2[6] — and metformin itself raises GLP-1 concentrations and alters the intestinal bile acid pool and microbiome[104]. The endogenous background in these populations is pharmacologically modified in the direction opposite to the hypothesised deficit. Third, drug exposure is not reduced in T2D: oral semaglutide bioavailability did not differ significantly between healthy participants and people with T2D, and subcutaneous exposure varies mainly with body weight[43,105].

The honest position is that the peripheral axis is a well-motivated, mechanistically detailed and almost entirely untested contributor. Its strongest form is not that agonists work less well on diabetic receptors — that version is weak and partly refuted. It is that T2D degrades a shared, multi-axis afferent pathway through which satiation signals of every kind must pass, removing a physiological component that normally operates alongside the drug, and that this loss is graded with glycaemic control. That proposition has never been measured in either population receiving an incretin agent. Section 12 sets out the studies that would test it, including the two most direct: whether baseline autonomic function or measured endogenous incretin secretion predicts weight response to an agonist, and whether the response differs in people with established vagal dysfunction.

6. Gastrointestinal physiology and gastric emptying

GLP-1 slows gastric emptying acutely. With sustained receptor exposure this effect diminishes. Tachyphylaxis of the delay was evident by a second meal during an 8.5-hour GLP-1 infusion, together with loss of the associated vagal marker pancreatic polypeptide[106]. A continuous 24-hour infusion attenuated the delay, whereas intermittent exposure preserved it[107]. With long-acting agents the picture is mixed. After 20 weeks of semaglutide 2.4 mg, paracetamol absorption showed no delay[56]. Semaglutide 1.0 mg delayed only first-hour emptying[108]. Exenatide once weekly still slowed scintigraphically measured emptying of solids and liquids at 8 weeks[109]. Liraglutide 3.0 mg delayed solid emptying at 5 and 16 weeks in obesity, and the delay at 5 weeks correlated with weight loss at 16 weeks[110]. Tirzepatide's delay of gastric emptying likewise waned with repeated dosing in healthy participants. In T2D, however, a residual delay persisted during dose escalation, and a long-acting GIP analogue alone did not affect emptying[111].

Diabetes affects this axis in ways that could push in either direction. Acute hyperglycaemia slows gastric emptying, as shown in type 1 diabetes under glucose clamp[112]. Diabetic autonomic neuropathy can cause gastroparesis[113]. The direction is not uniform, however, and the common assumption that diabetes means slow emptying is wrong as often as it is right. Among 108 patients with diabetes referred for upper gastrointestinal symptoms and tested by scintigraphy, emptying was abnormal in 56% but *rapid* in 37% and slow in only 19%; 40.3% of those with T2D had accelerated emptying at one hour[114]. Baseline emptying rate in a T2D trial population is therefore likely to be heterogeneous in both directions, which blunts any prediction that rests on it being uniformly delayed. The glucose-lowering effect of GLP-1 receptor agonism through slower gastric emptying depends on the baseline emptying rate[115,116]. If emptying is already slow in some people with T2D, the incremental slowing a drug can produce, and any satiation it adds, may be smaller. Conversely, correcting hyperglycaemia could speed emptying, partly offsetting the drug effect. Neither possibility has been tested against weight outcomes.

The weight of evidence suggests that gastric emptying contributes to early satiation and perhaps to variation between individuals in response[110]. It is not the principal mechanism of long-term weight loss, because weight loss continues after the delay has largely waned, and central receptor populations are sufficient for anorexia in animal models[59,60]. Gastrointestinal changes in T2D are therefore best regarded as a variable and probably minor modifier. They would need to be measured, together with weight, in both populations before they could be assigned any share of the gap. Tolerability, the clinically visible face of gut effects, does not appear worse in T2D trials (Section 2.6).

7. GIP biology and dual receptor agonism

7.1 Established human pharmacology

The insulinotropic action of GIP is reduced in T2D, both in peak effect[49] and in amplifying late-phase secretion[50]. Near-normalisation of glycaemia partly restores it[48]. In rat islets, chronic hyperglycaemia reduces expression of both GLP-1 and GIP receptors, and normalising glucose reverses the change[82]. This finding suggests a mechanism, but it is animal data. GIP has adipose actions in humans. Together with hyperinsulinaemia it increased subcutaneous adipose blood flow, glucose uptake, triglyceride hydrolysis and fatty-acid re-esterification[117]. Infused GIP did not reduce energy intake in overweight men, and adding GIP to GLP-1 did not enhance the GLP-1-induced reduction in intake[118].

7.2 Mechanisms proposed mainly from animal and cellular studies

In mice, hypothalamic cells expressing the GIP receptor suppress food intake when activated[119]. Central GIP receptors are required for the anorectic effect of acyl-GIP and for the greater efficacy of GLP-1/GIP co-agonism over GLP-1 alone[120]. In mice, rats and shrews, GIP receptor agonism attenuates GLP-1-induced emesis while preserving reduced food intake[121]. Paradoxically, GIP receptor antagonist antibodies also reduce weight and enhance GLP-1 agonist effects in rodents and non-human primates[122]. One explanation is that sustained agonism produces functional antagonism through receptor desensitisation, but this remains unresolved. At the receptor level, tirzepatide is biased in its signalling and engages the GIP receptor more than the GLP-1 receptor[54].

7.3 Does altered GIP physiology in T2D predict a different response to dual agonism?

If reduced GIP receptor responsiveness in T2D blunted the extra weight loss attributable to GIP agonism, the ratio of T2D to non-T2D weight loss should be lower for tirzepatide than for selective GLP-1 receptor agonists. It is not: the ratio was 0.59–0.65 for tirzepatide and 0.50–0.60 for semaglutide (Table 1). The attenuation is also present with maridebart cafraglutide, which *antagonises* the GIP receptor (ratio about 0.77)[25], and with orforglipron, which has no GIP activity[8]. The weight deficit in T2D therefore does not depend on the direction of GIP receptor pharmacology. This makes impaired GIP signalling an unlikely primary explanation.

The mechanistic data in T2D point instead to effects of tirzepatide beyond intake. In T2D, tirzepatide produced more weight and fat-mass loss than semaglutide 1 mg without a measurable difference in ad libitum intake[66]. It also produced larger improvements in insulin sensitivity[52]. In people with obesity but without diabetes, tirzepatide increased fat oxidation without changing metabolic adaptation[123]. Larger weight loss with dual agonism does not prove a specific GIP-mediated anorectic mechanism in humans. The relative contributions of greater GLP-1 receptor exposure, GIP-mediated tolerability, adipose effects and substrate effects are not established.

8. Insulin resistance, hyperinsulinaemia and substrate partitioning

8.1 The pathophysiological premise

Skeletal muscle is the main site of insulin-stimulated glucose disposal[124]. In T2D, insulin resistance affects muscle, liver and adipose tissue. Hepatic insulin resistance sustains glucose production, adipose insulin resistance weakens insulin's suppression of lipolysis, and ectopic lipid accumulation links these defects[125]. Insulin-resistant muscle shows "metabolic inflexibility": it oxidises more glucose in the fasting state but fails to switch towards glucose oxidation when stimulated by insulin[126]. Incretin

therapy improves several of these defects. Tirzepatide increased the clamp M value in T2D[52]. In patients with non-alcoholic steatohepatitis, liraglutide improved hepatic and adipose insulin sensitivity and reduced de novo lipogenesis[127]. Semaglutide reduced HOMA-IR in people with prediabetes[30].

8.2 Does insulin resistance plausibly limit weight loss?

A recurrent proposal is that insulin resistance itself, or the hyperinsulinaemia it causes, traps energy in adipose tissue or limits fat mobilisation and so slows weight loss. Three considerations argue against assigning it a primary role.

1. **Substrate handling does not set the energy balance.** Muscle glucose uptake, glucose oxidation and glycogen storage describe which fuel is used and where it is stored. Total energy expenditure is set by ATP demand. A shift from glucose to fat oxidation, or the reverse, does not by itself change the energy deficit. In a controlled metabolic-ward study, isocaloric carbohydrate restriction lowered insulin and raised fat oxidation, yet produced less body-fat loss than equal-calorie fat restriction[128]. Insulin-driven partitioning therefore did not override energy balance.
2. **The mechanistic link is unmeasured.** We found no human study in T2D that measured muscle glucose uptake, glucose oxidation, glycogen storage and total energy expenditure together during incretin treatment. The proposition that impaired insulin-mediated glucose disposal reduces incretin-associated weight loss is therefore untested.
3. **Treatment improves insulin sensitivity.** Insulin sensitivity improves during treatment in both populations, which would narrow any partitioning difference over time.

One partitioning effect deserves note because it changes weight without changing fat: glycogen. Glycogen is stored with water, so a larger glycogen pool can add weight. Improved insulin action could plausibly restore muscle and liver glycogen that are depleted in poorly controlled diabetes and add weight unrelated to adiposity. This is an inference from general physiology and has not been measured in incretin trials.

8.3 Endogenous hyperinsulinaemia versus insulin and sulfonylurea therapy

Endogenous hyperinsulinaemia must be distinguished from weight-promoting medicines. In UKPDS 33, intensive treatment with insulin increased weight by 4.0 kg, and sulfonylureas by 1.7–2.6 kg, relative to conventional therapy[129]. In SURMOUNT-2, 26.7% of participants took sulfonylureas, and level-2 hypoglycaemia clustered in that group despite protocol-guided dose reduction[6]. Insulin-associated weight gain arises partly through the energetics of improved glycaemia (Section 9). Hypoglycaemia-avoidant eating is often invoked but is a hypothesis. Notably, liraglutide 3.0 mg plus intensive behavioural therapy produced a placebo-adjusted reduction of 4.3 percentage points in insulin-treated T2D[130], similar to the 4.0 points in T2D without insulin[14]. Insulin co-therapy with down-titration therefore did not abolish the response. Background therapy is a credible contributor to the gap at population level, but no trial has stratified weight response by background agent with adequate power.

9. Glycosuria, glycaemic correction and net energy balance

9.1 The premise

When plasma glucose exceeds the renal reabsorptive capacity, glucose is lost in urine and its energy leaves the body. In people with T2D under stepped hyperglycaemic clamps, urinary glucose excretion rose from 0.06 mmol/min at 7.8 mmol/L to 0.77 mmol/min at 13.3 mmol/L. Glycosuria persisted down to 7.8 mmol/L, and the threshold varied between people from 7.7 to 12.2 mmol/L[131]. Restoring glycaemia closes this energy leak. The energy is then retained, acting like an increase in intake, but only in people who were glycosuric, and only in the arm whose glycaemia improves.

9.2 Measured magnitudes

Three human studies measured the energy involved directly.

- In obese Pima Indians with T2D, 6 weeks of sulfonylurea lowered fasting glucose from 253 to 144 mg/dL and urinary glucose from 128 to 11 g/day. Resting metabolic rate per kilogram of fat-free mass also fell. To keep weight constant, daily intake had to be cut from 3070 to 2784 kcal[132].
- In type 1 diabetes, intensive insulin therapy under constant intake reduced glycosuria from 428 to 39 mmol/day. Whole-room calorimetry showed a 5% (118 kcal/day) fall in 24-hour energy expenditure, and weight increased by 2.6 kg, almost all of it fat[133].
- In insulin-treated T2D, improved glycaemia was an independent determinant of lower basal metabolic rate. The authors concluded that better glycaemic control promotes weight gain by reducing both metabolic rate and glucosuria[134].

The reverse experiment confirms that urinary glucose energy matters to weight. SGLT2 inhibition with empagliflozin caused about 206 kcal/day of glycosuric loss, which predicted an 11.3 kg weight loss. Observed loss was 3.2 kg because calculated intake rose by about 13%[135]. With canagliflozin, intake rose by about 100 kcal/day for each kilogram lost[74].

9.3 A formal model: Hall 2026

The most direct quantitative treatment of this question is a modelling study that appeared while this review was in preparation[41]. It is the closest existing approach to proposed study 2 in Table 4, conducted *in silico* rather than by measurement.

The design is the important part. A previously validated model of human energy balance dynamics[136] was fitted to the semaglutide response of participants *without* diabetes in STEP UP[11], estimating the parameters that describe how semaglutide affects appetite and energy intake. Those appetite parameters were then **held fixed** and the model applied to STEP UP T2D[12], adding only two diabetes-specific terms:

- **Urinary glucose excretion**, computed from estimated glomerular filtration rate, the renal threshold for glucose reabsorption[137] and plasma glucose derived from measured HbA1c via the ADAG equations[138]. Mean eGFR in STEP UP T2D was 92 mL/min/1.73 m², so a normal renal threshold of 180 mg/dL was assumed.
- **Energy expenditure**, reduced by 3.5 kcal/day per mg/dL fall in fasting plasma glucose, a coefficient derived from cross-sectional data in which resting energy expenditure in obese T2D was better predicted when fasting glucose was included, and in which poor glycaemic control corresponded to an REE elevation of up to 8%[139].

Because appetite parameters were not refitted, any weight-loss gap the model reproduces is attributable to those two energetic terms alone, and any residual is not. The reported result is that the attenuation in patients not on background SGLT2 inhibitor therapy was **partially explained** by the UGE and expenditure changes[41]. The author's conclusion is that these effects "can substantially offset negative energy balance," and that they "may partially explain" the gap — not that they account for it.

The SGLT2 inhibitor prediction. The model's most striking and counterintuitive output is that patients on *stable background* SGLT2 inhibitor therapy should show the **greatest** attenuation of weight loss. The logic is that an SGLT2 inhibitor lowers the renal threshold (modelled as 110 mg/dL), raising baseline UGE by roughly 100 g/day. Starting from that elevated baseline, semaglutide-induced glycaemic improvement produces a correspondingly larger fall in UGE, and so a larger energy offset. Crucially, the model predicts the opposite for *simultaneous initiation* of an SGLT2 inhibitor and an incretin agent, which should yield greater weight loss than either alone. Stable background therapy and co-initiation therefore move weight response in opposite directions — a distinction with direct implications for trial design and for interpreting the roughly 30% of STEP UP T2D participants who were on background SGLT2 inhibitors.

What the model does not settle. It is a simulation, not a measurement, and a preprint that has not been peer reviewed. The study is funded by AstraZeneca and the author is an employee of that company[41]; this does not invalidate the analysis, whose assumptions are stated explicitly and are independently checkable, but it should be recorded. Its own stated limitations are substantial and align closely with the gaps identified elsewhere in this review: aggregate trial means rather than individual participant data; the

renal threshold estimated from the literature rather than measured; real uncertainty about the magnitude and variability of the expenditure effect; sensitivity to the choice of HbA1c-to-average-glucose equation, with DCCT-derived equations predicting substantially greater baseline UGE and hence greater attenuation; HbA1c being a lagging indicator, which may distort early trajectories; the trial not reporting results separately by background SGLT2 inhibitor status, so the two simulations were averaged 30:70; and a higher proportion of women in STEP UP than in STEP UP T2D (~74% vs ~52%), a known modifier of weight response. The author's own recommendation is that future studies directly measure dynamic changes in UGE and energy expenditure to test the predictions.

One discrepancy should be flagged. The weight losses quoted in the preprint for STEP UP (approximately 18% and 21% at 72 weeks for 2.4 mg and 7.2 mg) and STEP UP T2D (approximately 11% and 14%) exceed the treatment-policy values in the primary publications (−15.6% and −18.7%; −13.2% for 7.2 mg)[11,12]. This is consistent with the model having been fitted to a trial-product or on-treatment estimand, but the preprint does not state which estimand was simulated. Table 1 uses the published treatment-policy values throughout.

9.4 Our own scenario estimates

None of the incretin trials measured urinary glucose or energy expenditure directly. Independently of the model above, we built transparent scenario estimates to bound the effect from measured inputs (Table 2; analysis/derived.py). Each assumes a *differential* reduction in glycosuria between drug and placebo arms, plus a differential fall in resting energy expenditure, persisting over the trial. We used 3.75 kcal per gram of glucose and the energy–weight relationship of Hall and colleagues: a permanent 100 kJ/day change yields about 1 kg at steady state, with half the change reached in about 1 year[140]. For trial-like T2D populations with baseline HbA1c of about 8%, in which the drug lowers HbA1c by about 1.5 percentage points more than placebo[6], the retained energy could account for roughly 1 kg (low scenario) to 6–7 kg (high scenario) by 68–72 weeks. The observed placebo-adjusted gap between STEP 1 and STEP 2 is about 6 percentage points, or roughly 6 kg.

These calculations show that the energetics of glycaemic correction are *quantitatively capable* of explaining a meaningful share of the gap. They do not show that they do. Four caveats apply. Baseline glycosuria in trial populations was not measured and varies widely between individuals[131]. Behavioural compensation could be blunted by the drug's own anorectic effect, or it could add to the offset. The anchoring studies of metabolic rate involved poorer control than in modern trials[132]. And the gradient seen in prediabetes, where glycosuria is negligible (Section 2.4), shows that glycosuria cannot be the whole explanation.

Our scenarios and the formal model agree on the central point and differ usefully in method. Both conclude that the energetics of glycaemic correction can account for a substantial but partial share of the gap. The model adds physiological structure our scenarios lack — a renal-threshold-based UGE calculation, an explicit expenditure coefficient, and dynamic rather than steady-state weight prediction — and it generates a testable prediction about background SGLT2 inhibition that scenario arithmetic could not reach. Neither rests on a single measurement of urinary glucose or energy expenditure made during incretin treatment, because none exists.

9.5 What the mechanism explains well, and what it does not

The mechanism fits three features of the clinical phenotype. It operates only in the arm whose glycaemia improves, so it predicts a deficit in drug-attributable rather than placebo-arm weight change, as observed — most cleanly in STEP UP and STEP UP T2D, where placebo-arm change was −3.9% in both[11,12]. Because people without diabetes have no glycosuria to lose, it predicts no offset in non-diabetes trials. And because the offset scales with the magnitude of glycaemic correction, it predicts a gradient with baseline HbA1c, which is what the within-T2D analyses show[28,29].

It is also consistent with the gap appearing outside incretin pharmacology altogether (Section 2.5): any intervention that lowers weight and corrects glycaemia in T2D incurs the same energetic offset, whether it acts on the GLP-1 receptor, the 5-HT2C receptor, or the stomach[32,34,37,38].

Two things it does not explain. The prediabetes step, where glycosuria is negligible yet roughly 2 percentage points of deficit have already appeared (Section 2.4). And the residual that the model itself leaves unexplained, which the author describes as partial explanation rather than full[41]. Something further is operating, and Sections 5 to 8 and 10 set out the candidates.

Table 2. Energy offsets from glycaemic correction: measured anchors and assumption-driven scenarios.

Source / scenario	Input	Energy (kcal/day)	Weight equivalent
Bogardus 1986, T2D, sulfonylurea[132]	Urinary glucose 128 → 11 g/day (measured)	~439	—
Carlson & Campbell 1993, type 1 diabetes, intensive insulin[133]	Glycosuria 428 → 39 mmol/day; 24-h EE −118 kcal/day (measured)	~263 (glycosuria) + 118 (EE)	+2.6 kg in 2 months at constant intake
Ferrannini 2015, empagliflozin[135]	Glycosuric loss (estimated from measurements)	~206	Predicted −11.3 kg; observed −3.2 kg (compensation)
Rave 2006 clamp[131]	UGE 0.06 → 0.77 mmol/min at 7.8 → 13.3 mmol/L	58–749 if sustained for 24 h	—
Gougeon 2002, obese T2D (n=65)[139]	REE elevation with poor glycaemic control (measured, cross-sectional)	up to +8% of REE; modelled as 3.5 kcal/day per mg/dL fasting glucose[41]	—
Hall 2026 model, T2D without background SGLT2i[41]	UGE from eGFR, renal threshold 180 mg/dL and HbA1c-derived glucose, plus expenditure term	not stated as a single value	Partially explains the STEP UP T2D gap
Hall 2026 model, T2D on stable background SGLT2i[41]	Renal threshold lowered to 110 mg/dL; baseline UGE ~100 g/day higher	~375 (at 3.75 kcal/g)	Greatest predicted attenuation of all groups modelled
Scenario: low	10 g/day differential glycosuria; no REE change	38	~1.0 kg by 68–72 weeks
Scenario: central	25 g/day; REE −40 kcal/day	134	~3.3 kg by 68–72 weeks
Scenario: high	45 g/day; REE −90 kcal/day	259	~6.4–6.7 kg by 68–72 weeks

Scenarios are illustrative assumptions, not measurements. Conversion: 23.9 kcal/day per kg at steady state; exponential approach with a 1-year half-time[140].

10. Adipose tissue, body composition and metabolic adaptation

Composition of weight lost. In a DXA substudy of SURMOUNT-1 (no diabetes), tirzepatide reduced weight by 21.3%, fat mass by 33.9% and lean mass by 10.9%. About three-quarters of the weight lost was fat in both drug and placebo groups[141]. In T2D, semaglutide 1.0 mg reduced fat mass by 3.4 kg and lean mass by 2.3 kg over 52 weeks in a SUSTAIN 8 substudy, but there was no placebo arm and the dose was lower[142]. Tirzepatide reduced liver fat and visceral and subcutaneous adipose volumes compared with insulin degludec[143]. Semaglutide 2.4 mg reduced visceral fat area by 40% in an East Asian population with and without T2D[144]. No study has compared body-composition changes between populations with and without T2D at matched dose and duration.

Body composition cannot easily explain smaller weight loss. Losing a kilogram of fat removes about 39.5 MJ of stored energy, compared with about 7.6 MJ for a kilogram of lean tissue[140]. If people with T2D lost a larger share of weight as lean tissue, a given energy deficit would produce *more* weight loss, not less. A composition difference could reduce weight loss only through fluid or glycogen shifts (Section 8.2). Those are short-term effects and have not been measured in this context.

Energy expenditure and adaptation. In humans, incretin agonists do not increase energy expenditure. Semaglutide did not change resting metabolic rate adjusted for lean mass[55]. Liraglutide did not change 24-hour expenditure in T2D[53] and transiently *reduced* resting expenditure in a 26-week T2D trial[145]. Tirzepatide did not change metabolic adaptation in obesity but increased fat oxidation[123]. The GLP-1/glucagon co-agonist cotadutide produced weight loss in T2D driven mainly by reduced intake, with little

change in expenditure[146]. One adaptation is specific to diabetes. Resting metabolic rate is about 5% higher in T2D than in equally obese people with normal glucose tolerance, and it falls as glycaemia is corrected[132,134]. Normalising glycaemia thus removes an energy cost unique to the diabetic state, adding to the glycosuric offset in Section 9.

Physical activity has not been measured objectively and compared between populations in incretin trials. Whether lower activity in older, more comorbid T2D populations contributes is unknown.

11. Integrative model: from diabetes pathophysiology to weight-loss response

Figure 2 organises the evidence into four interacting domains. Each causal link is labelled by the strength of its support. Table 3 summarises the evidence for each candidate mechanism.

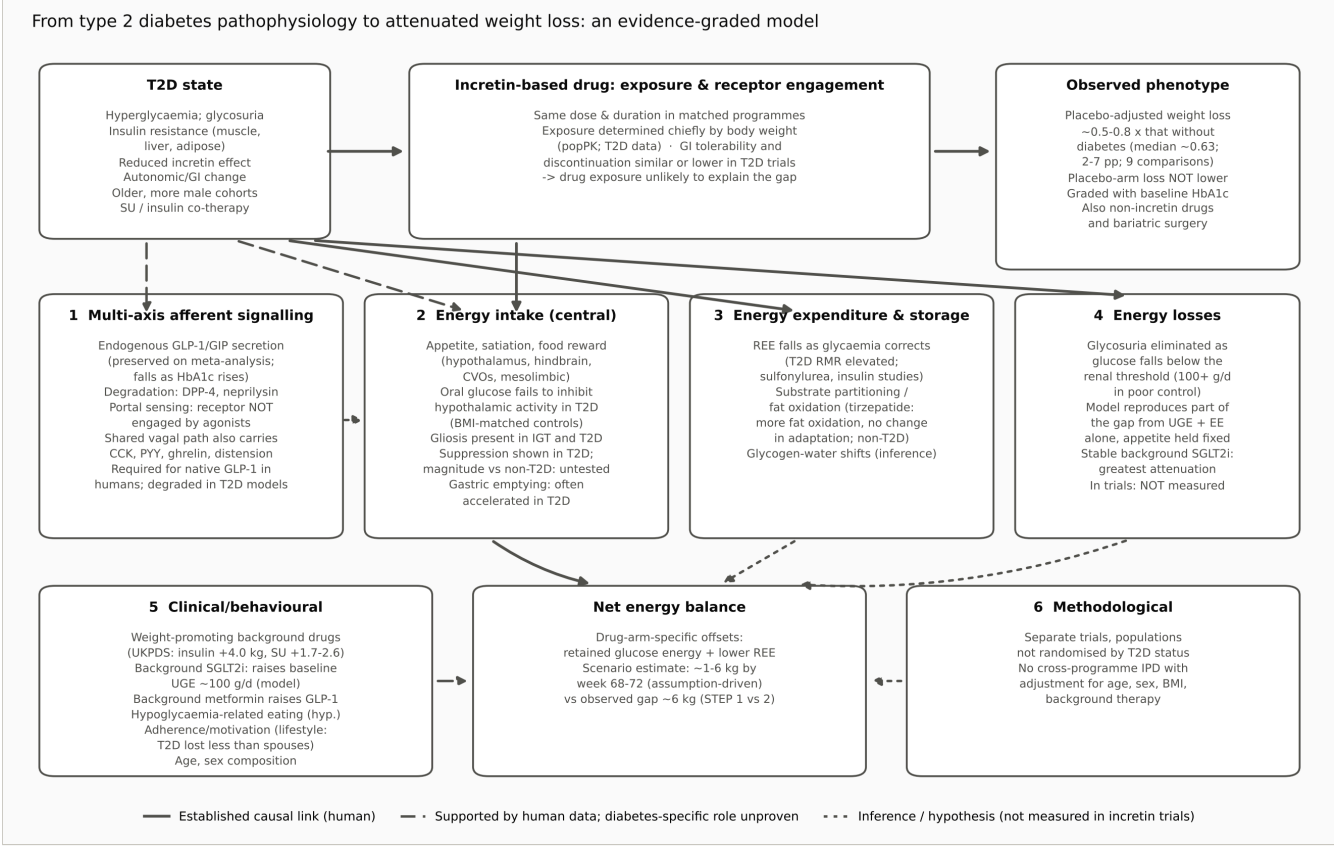


Fig. 2 | Evidence-graded integrative model. Features of T2D act on the peripheral incretin signalling apparatus and on three components of energy balance (intake, expenditure and storage, and losses), alongside clinical and behavioural factors and methodological sources of between-trial difference. The peripheral domain (box 1) is placed upstream of central intake regulation (box 2), to which it is linked by a dotted arrow: the proposition that degraded peripheral signalling reduces the drug's anorectic effect in T2D is a hypothesis that has not been measured in either population on treatment. Solid lines denote causal links established in humans. Dashed lines denote links supported by human data whose diabetes-specific role is unproven. Dotted lines denote inferences or hypotheses not measured in incretin trials. Scenario values are assumption-driven (Table 2). The figure does not apportion the gap among mechanisms.

Drug exposure and pharmacodynamics. Dose, duration and estimand were matched in the comparisons. Exposure depends mainly on body weight, and tolerability and discontinuation do not favour the trials without diabetes. Receptor pharmacology varies widely across agents that all show the deficit, including opposite directions of GIP receptor action. Reduced drug exposure or a receptor-specific defect is therefore unlikely to be a primary determinant.

Energy intake. Intake suppression is the main mechanism of incretin-induced weight loss in both populations and is preserved, at least qualitatively, in T2D. Its magnitude has never been compared between populations under matched conditions. A smaller central or gastrointestinal effect in T2D, through hypothalamic gliosis, central insulin resistance, greater food-cue reactivity or altered baseline gastric emptying, is biologically plausible but unmeasured. This is the largest unknown in the model.

Peripheral incretin signalling. Upstream of the brain sits a secretory, degradative and afferent apparatus that diabetes measurably alters (Section 5). The gut–vagal glucose-sensing pathway degrades in type 2 diabetic and insulin-resistant animals, intact vagal innervation is required for native GLP-1 to suppress intake in humans, and agonist-induced anorexia is blunted in a dysmetabolic rodent host with nodose receptor downregulation. Three findings bound the contribution: incretin secretion is on average preserved in T2D and falls only with worsening glycaemic control; the portal afferent receptor appears not to be engaged by exogenous agonists; and prolonging native GLP-1 with DPP-4 inhibition does not reduce weight. The most defensible version of this mechanism is therefore not pharmacological resistance but the loss of a physiological, paracrine satiation signal that normally acts alongside the drug — graded with glycaemic control, and never measured in either population on treatment.

Energy expenditure and substrate metabolism. Agonists do not raise expenditure, and adaptive changes appear similar in kind across populations. Correcting hyperglycaemia removes the excess resting metabolic rate seen in T2D. Changes in substrate partitioning and insulin sensitivity alter fuel selection, but there is no evidence that they set the size of the energy deficit.

Energy losses and glycaemic correction. Elimination of glycosuria returns previously lost glucose energy to the body, and correcting hyperglycaemia removes the excess resting metabolic rate of the diabetic state. The size of both effects is established in diabetes generally. A formal energy-balance model, fitted to non-diabetic semaglutide response and then applied to T2D with appetite parameters held fixed, partially reproduces the observed gap from these two terms alone, and predicts that stable background SGLT2 inhibition should produce the greatest attenuation[41]. Our independent scenario arithmetic reaches a compatible magnitude. This is now the best-supported mechanism in the model. Neither component has been measured in an incretin trial.

How several modest effects combine. Together, the energy offsets from glycaemic correction (glycosuria and metabolic rate) and weight-promoting co-therapies could narrow the net energy deficit in the drug arm. Because weight approaches a new steady state over a year or more, small persistent offsets matter: each 24 kcal/day of

persistent offset corresponds to about 1 kg at steady state[140]. Population differences in sex and age, and possibly behaviour[27,39], add further differences between trials. We deliberately do not assign percentage contributions. None of these components has been measured in both populations under the same treatment, so any apportionment would be speculative.

11.1 Three host-level hypotheses, and what would separate them

The cross-intervention evidence (Section 2.5) establishes that the operative mechanism is in the host, not in incretin receptor pharmacology. It does not say which host mechanism. Three remain live, they are not mutually exclusive, and the phenotype as currently described is compatible with all three:

- **H1 — Energetics.** Correcting hyperglycaemia returns glycosuric energy and removes the excess resting metabolic rate of the diabetic state, narrowing the net deficit in the drug arm (Section 9).
- **H2 — Generalised anorectic-responsiveness deficit.** The diabetic brain responds less to satiation signalling of any kind, so a given pharmacological or surgical stimulus produces less intake suppression (Section 4.3).
- **H3 — Degraded shared afferent machinery.** The multi-axis gut–brain pathway that carries CCK, PYY, ghrelin, distension and incretin signals alike is damaged, blunting transmission regardless of which signal is amplified (Section 5).

The table below sets out where they diverge. The essential point is that almost every observation currently in hand is predicted by all three; only two rows discriminate, and both are thinly populated.

Observation	H1 energetics	H2 appetite response	H3 afferent machinery
Deficit confined to the drug-attributable component	Predicted	Predicted	Predicted
Graded with baseline HbA1c	Predicted	Predicted if graded with dysglycaemia	Predicted
Appears with non-incretin anorectics and after surgery	Predicted	Predicted	Predicted
Ratio similar across receptor pharmacologies	Predicted	Predicted	Predicted
Deficit already present in prediabetes	**Not predicted** (glycosuria negligible)	**Predicted** (gliosis present in IGT)	**Predicted** (afferent changes precede overt diabetes)
Gap persists under fixed, enforced energy intake	**Predicted**	**Not predicted**	**Not predicted**
Measured UGE + REE change accounts for the deficit	Predicted	Not predicted	Not predicted
Placebo-adjusted intake reduction smaller in T2D	Not predicted	Predicted	Predicted
Weight response tracks autonomic function independently of HbA1c	Not predicted	Not specifically	Predicted

Two rows carry the weight. The **prediabetes step** favours H2 and H3: roughly 2 percentage points of deficit have appeared where there is almost no glycosuria to retain, and the central and afferent abnormalities are documented in impaired glucose tolerance[71,99]. The **fixed-intake experiment** favours H1: when intake was enforced at 300 kcal/day, the gap persisted[38] — though with a baseline composition confounder, and against a meta-analytic null[40]. These two rows point in opposite directions, which is the honest summary of the field.

One methodological point deserves emphasis because it is easy to misread. The modelling analysis of Section 9.3 held the appetite parameters **fixed** at their non-diabetic values[41]. That is what makes its energetic attribution clean, but it also means the model was never allowed to express H2. A blunted appetite response in T2D could not appear as a fitted parameter; it could only appear as residual unexplained gap — which is exactly what the analysis reports. The model therefore demonstrates that energetics can account for *part* of the gap. It cannot distinguish "energetics explains part and nothing else is needed" from "energetics explains part and blunted anorectic response explains the rest," because the second possibility was excluded by construction. Refitting the appetite parameters in T2D and reporting how much they must change would be a direct and inexpensive test.

The single most discriminating measurement is therefore energy intake. If placebo-adjusted intake suppression is equal in both populations on the same agent and the weight gap persists, H1 is right and H2 is wrong. If intake suppression is smaller in T2D, H2 or H3 is operating and energetics alone cannot be the answer. No study has made this comparison (Section 4.2), which is why it is the first study in Table 4.

Which explanations are best supported? Ranked by evidence:

- **Best supported (the phenotype itself).** The gap exists, is graded with glycaemic status, is confined to the drug-attributable component, and is not specific to incretin drugs — it appears with unrelated pharmacotherapies and after bariatric surgery. Drug exposure and tolerability do not explain it. The mechanism is therefore in the host.
- **Best *quantified*, and the only one with a formal model: energetics (H1).** Glycosuric energy retention and the fall in resting expenditure are established in other settings, are quantitatively capable of a substantial share, and partially reproduce the gap in simulation[41]. This is the best-*characterised* hypothesis, which is not the same as the best-*supported*: its mediation has never been measured, it cannot explain the prediabetes step, and the model that supports it excluded the main competing hypothesis by construction.
- **Equally compatible with the evidence, far less investigated: blunted anorectic responsiveness (H2) and degraded shared afferent machinery (H3).** Both predict everything H1 predicts, plus the prediabetes step that H1 cannot. Both have direct human mechanistic support — failed hypothalamic inhibition after glucose in BMI-matched T2D[70], gliosis in IGT and T2D[71], loss of GLP-1's intake effect after human vagotomy[94] — and neither has been tested as a mediator of the weight gap. Their lower ranking in earlier drafts reflected the absence of a quantitative model, not the weight of evidence against them.
- **Plausible and quantitatively capable, but unmeasured.** Weight-promoting background therapy. Background SGLT2 inhibition. Population composition by sex and age.
- **Weakest.** An intrinsic, insulin-resistance-driven pharmacological resistance. Impaired GIP signalling as the primary cause. A reduced endogenous incretin *secretory* deficit as the primary cause. Note these are weak as *primary* explanations; the broad, multi-axis version of the afferent hypothesis (H3) is not among them.

Table 3 | Mechanistic evidence map.

Mechanism (category)	Biological hypothesis	Evidence in T2D	Evidence without diabetes	Direct comparison by diabetes status	Human vs animal	Causal evidence	Quantitative contribution	Confidence
Reduced drug exposure / tolerability (clinical)	Lower exposure or more discontinuation in T2D	Exposure set by body weight[43,44]; GI events and discontinuation not higher[8,10]	Same[7,9]	Indirect (matched programmes)	Human	Argues against	Not a major contributor	Moderate (against)
Attenuated appetite suppression (direct)	Diabetic state blunts central/vagal anorectic response	Suppression present[64–67]	Suppression present[55–57]	None	Human (effect); animal (circuits)[59,60]	Not tested	Unknown	Low

Mechanism (category)	Biological hypothesis	Evidence in T2D	Evidence without diabetes	Direct comparison by diabetes status	Human vs animal	Causal evidence	Quantitative contribution	Confidence
Central gliosis / brain insulin resistance (direct)	Hypothalamic injury or insulin resistance lowers drug efficacy	Gliosis greater in IGT/T2D[71]; brain IR[72]	—	None	Human association	None	Unknown	Very low
Impaired central nutrient sensing (H2, direct)	The diabetic hypothalamus fails to register ingested nutrient, blunting satiation from any source	Glucose ingestion fails to inhibit hypothalamic activity in T2D vs BMI-matched controls[70]; altered food-cue processing[67,73]	Inhibition present in healthy controls[70]	None on treatment	Human	Not tested as a mediator	Unknown	Low–moderate (mechanism); none (mediation)
Multi-axis satiety signalling (H3, direct)	Shared afferent path carrying CCK, PYY, ghrelin and distension signals is degraded, blunting all anorectic inputs	Roles of ghrelin/CCK/GLP-1/PYY in eating in obesity and post-bypass are uncertain[102]; bile acid–CCK–GLP-1 axis interlinked and perturbable in T2D[103]; afferent activation impaired in T2D models[99]	CCK established as an endogenous control of eating at healthy weight[102]	None	Human + animal	Not tested	Unknown	Low (untested)
Reduced endogenous incretin secretion (direct/indirect)	Lower native GLP-1 removes a paracrine signal acting alongside the drug	Individual studies show reduced GLP-1[78,79]; meta-analyses show secretion preserved overall, reduced only with higher HbA1c/FPG[80,81]	Preserved	Not measured on treatment	Human	Argues against a uniform deficit; graded effect possible	Unknown; bounded by DPP-4-inhibitor weight neutrality	Low
Increased incretin degradation (DPP-4, neprilysin) (direct)	Higher degradation lowers active GLP-1 at paracrine and portal receptors	DPP-4 activity correlates with HbA1c and HOMA-IR within T2D[83,84]; gut rather than plasma DPP-4 drives the incretin effect[85]; neprilysin also degrades GLP-1[86]	—	None	Human association; rodent mechanism	None for weight	Small; DPP-4 inhibition raises GLP-1 without weight loss	Very low
Portal / hepatoportal GLP-1 sensing (direct)	Loss of portal GLP-1-receptor signalling blunts satiation	Portal receptor density and afferent responsiveness reduced in insulin resistance[100]; enteric and vagal activation impaired in T2D rats[99]	Portal sensor intact[87,88]	None	Animal	Argues against relevance to agonists: portal afferent receptor not engaged by exendin-4[92] or exenatide[93]; hepatic branch not required[89]	Probably small for drug effect	Low
Vagal afferent transmission / autonomic neuropathy (direct)	Diabetic vagal damage degrades the route carrying the anorectic signal	CAN in 18.6% at diagnosis[95,96]; vagal fibre loss (type 1, n=5)[97]; db/db nerve changes, vagus late[98]; agonist anorexia blunted with nodose GLP-1R downregulation[101]	GLP-1 intake suppression lost after human truncal vagotomy[94]; but circulating GLP-1 acts non-vagally[90]; central receptors sufficient[59,60]	None	Human (vagotomy); animal (diabetes models)	Not tested in T2D on treatment	Unknown	Low
Background metformin (clinical)	Metformin raises endogenous GLP-1 and alters bile acids and microbiome, modifying the background	Metformin in 89% of SURMOUNT-2[6]; raises GLP-1, bile acid pool, microbiome[104]	Absent	None	Human	Acts opposite to a secretory-deficit hypothesis	Unknown	Low
Gastric emptying (direct/indirect)	Slower baseline GE or neuropathy limits incremental slowing	GE effect baseline-dependent[115,116]; residual delay with tirzepatide in T2D[111]; hyperglycaemia slows GE[112]	Delay wanes; correlates with weight loss[56,110]	None	Human	Not tested	Probably minor	Low
Impaired GIP signalling (direct)	Reduced GIPR response blunts tirzepatide-specific weight loss	GIP insulinotropic action reduced[49,50]; partly restored[48]	GIP does not reduce intake in humans[118]	Ratios similar across GIPR agonist, antagonist and GLP-1-only agents (Table 1)	Human + animal[120,122]	Argues against primary role	Small at most	Moderate (against)
Glycosuria elimination (indirect)	Retained glucose energy offsets the deficit in the drug arm	Large measured glycosuria in poor control[131,132]; SGLT2i energetics[135]; formal model partially reproduces the STEP UP T2D gap with appetite parameters fixed[41]	Absent	Modelled, not measured, in incretin trials	Human + model	Established energetics; mediation untested	Scenario ~1–6 kg; model: partial explanation	Moderate–high (plausibility); low (mediation)

Mechanism (category)	Biological hypothesis	Evidence in T2D	Evidence without diabetes	Direct comparison by diabetes status	Human vs animal	Causal evidence	Quantitative contribution	Confidence
REE fall with glycaemic correction (indirect)	Excess REE of T2D removed	RMR +5% in T2D, falls with treatment[132,134]; REE up to +8% with poor control[139]; EE −5% with intensive insulin (T1D)[133]	Not applicable	Modelled as 3.5 kcal/day per mg/dL fasting glucose[41]; not measured	Human + model	Established in other settings	~40–100 kcal/day	Moderate (plausibility)
Background SGLT2i (clinical/indirect)	Lowered renal threshold raises baseline UGE, so glycaemic correction removes more energy	Predicted greatest attenuation on stable background SGLT2i; greater weight loss on co-initiation[41]; renal threshold measurable[137]	Not applicable	Never stratified; STEP UP T2D did not report by SGLT2i status[12]	Model	Untested	~100 g/day UGE difference (modelled)	Low (untested prediction)
Non-incretin generality (clinical)	The gap reflects host energetics, not receptor pharmacology	Smaller weight loss in T2D with naltrexone–bupropion[32], lorcaserin[34], phentermine–topiramate[36], bariatric surgery[37], VLCD[38]	Larger responses[33,35]	None randomised; formula-diet meta-analysis null[40]	Human	Argues against receptor-level primary cause	Not quantifiable	Moderate
Insulin resistance / substrate partitioning (direct)	IR or hyperinsulinaemia limits fat mobilisation	IR improves on treatment[52,127]	Isocaloric partitioning does not override energy balance[128]	None	Human	None for weight	Unknown; probably small	Low
Weight-promoting co-therapy (clinical)	SU/insulin add energy retention or defensive eating	Weight gain with SU/insulin[129]; SU in 27%[6]; insulin-treated response preserved[130]	Absent	None stratified	Human	Plausible	Unknown	Low–moderate
Population composition (methodological)	Age, sex and comorbidity differ between programmes	Older, more men[6,25]	More women[3]	Sex interaction for weight[24]	Human	Confounding plausible	Unknown	Moderate (plausibility)
Behaviour / adherence (clinical)	Lower dietary adherence or motivation	T2D lost less with behavioural programme; smaller intake reduction[39]; motivation proposed[27]	—	Small direct study (n=12 pairs)	Human	Associational	Unknown	Low

12. Unresolved questions and future directions

The central gap is that no study has measured the components of energy balance in people with and without T2D receiving the same incretin agent. A formal modelling analysis has now shown that the energetic offsets of glycaemic correction are capable of partially explaining the gap, and has generated specific, falsifiable predictions[41] — but a model that reproduces an observation is not evidence that the modelled mechanism produced it, since competing mechanisms may fit the same trajectories equally well. Direct measurement remains the decisive step, and the author of that analysis says so. Table 4 sets out studies that would resolve the main uncertainties, in order of priority.

Table 4 | Proposed studies.

#	Hypothesis	Population	Key measurements	Primary endpoint	Supports the hypothesis if…	Refutes it if…
1	The gap persists after adjustment for population composition	IPD from STEP 1/2, SURMOUNT-1/2, ATTAIN-1/2, REDEFINE 1/2	Weight; age, sex, BMI, HbA1c, diabetes duration, background therapy, adherence	Adjusted diabetes × treatment interaction for % weight change	An adjusted interaction remains, graded by HbA1c	The interaction disappears after adjustment
2	Glycaemic correction offsets the energy deficit	Matched adults with obesity ± T2D starting the same agent; 26–52 weeks	24-h urinary glucose; doubly labelled water; whole-room calorimetry; DXA; measured renal threshold	Change in urinary energy loss plus REE change versus weight-loss difference	Retained energy accounts for a substantial share of the weight-loss difference, and measured values match the modelled ones[41]	Urinary and REE changes are small (<30 kcal/day)
3	Appetite suppression is smaller in T2D (the single most discriminating measurement; see Section 11.1)	Same cohort as #2, with an early crossover arm	Ad libitum intake (24 h), appetite ratings, food-cue fMRI, hypothalamic response to oral nutrient[70] at weeks 3 and 20	Placebo-adjusted reduction in intake by diabetes status	Smaller reduction in T2D, predicting weight response — energetics alone cannot be the answer	Equal reduction in intake, with the weight gap persisting — energetics is the answer
3b	The model's energetic attribution survives allowing appetite to differ	Reanalysis of the Hall model[41]	Refit the appetite parameters in T2D rather than holding them at non-diabetic values	Magnitude of the change in fitted appetite parameters	Appetite parameters must worsen substantially to fit T2D data, implying H2 contributes	Parameters barely change; energetic terms suffice
4	Insulin resistance modifies weight response	Same cohort, stratified by clamp M value	Hyperinsulinaemic–euglycaemic clamp; stable-isotope tracers of lipolysis and glucose disposal; muscle glycogen by ^{13}C-MRS	Association of baseline M value with weight response, adjusted for intake	M value predicts response independently of intake and glycosuria	No independent association
5	Background therapy contributes	T2D trial participants randomised to protocolised sulfonylurea/insulin withdrawal versus continuation	Weight; hypoglycaemia; intake	Difference in weight response	Withdrawal increases weight loss	No difference

#	Hypothesis	Population	Key measurements	Primary endpoint	Supports the hypothesis if…	Refutes it if…
6	Background SGLT2i status modifies the weight response, in opposite directions for stable background versus co-initiation	T2D participants starting the same incretin agent, stratified into (a) stable background SGLT2i ≥90 days, (b) SGLT2i co-initiated with the incretin, (c) no SGLT2i	24-h urinary glucose at baseline and on treatment; HbA1c; weight; intake	Placebo-adjusted % weight change by stratum	Stable background SGLT2i shows the *greatest* attenuation while co-initiation shows the least, as the model predicts[41]	Strata do not differ, or stable background SGLT2i shows *less* attenuation
7	Peripheral vagal integrity modifies weight response	Adults with obesity ± T2D starting the same agent, stratified by autonomic function	Cardiovascular autonomic reflex tests; gastric emptying (scintigraphy); heart rate variability; weight; ad libitum intake	Weight response by baseline autonomic function tertile, adjusted for HbA1c and BMI	Poorer vagal function predicts smaller weight loss independently of glycaemia	No association, or association explained by HbA1c
8	The endogenous incretin background predicts response	Existing phase 3 trial biobanks (STEP, SURMOUNT, ATTAIN) with stored samples	Baseline meal-stimulated total and intact GLP-1 and GIP; plasma DPP-4 activity; HbA1c	Association of baseline incretin response with placebo-adjusted % weight change	Lower endogenous secretion predicts smaller drug response	No association after adjustment for HbA1c
9	Vagal transmission is required for agonist-induced anorexia in humans	Adults with prior truncal vagotomy and matched controls, or T2D with versus without gastroparesis	Ad libitum intake and appetite ratings on a long-acting agonist; gastric emptying	Placebo-adjusted reduction in intake by vagal status	Reduced or absent intake suppression with vagal dysfunction	Equal suppression, locating the effect centrally

Several design principles apply across these studies:

- **Mediation analyses.** Studies should prespecify mediation analyses that partition the diabetes difference into intake, expenditure and urinary-loss components, rather than reporting each in isolation.
- **Controlled feeding.** Short controlled-feeding studies under matched agonist exposure could separate drug effects on intake from effects on energy disposal.
- **Measurement in T2D trials.** Phase 3 programmes in T2D could add urinary glucose collection and expenditure measurement at low cost, which would make these data routine. Baseline autonomic reflex testing and a stored meal-stimulated incretin sample would add similarly little burden and would make studies 7 and 8 answerable retrospectively.
- **Separating pathway from host.** Studies 7 and 9 test the peripheral hypothesis of Section 5 in the only way available in humans: by varying vagal integrity rather than diabetes status. Because vagal dysfunction and hyperglycaemia co-vary, either design needs adjustment for HbA1c, and a null result in study 9 would locate the agonist's anorectic effect centrally and largely close the question.

13. Conclusions

Clinical question. The evidence firmly supports an association between T2D and less weight loss on comparable incretin-based therapy. Across nine matched comparisons spanning six molecules, the placebo-adjusted reduction in T2D was about 50–80% (median ~63%) of that without diabetes, an absolute shortfall of about 2–7 percentage points at 52–72 weeks. The phenotype is not confined to incretin pharmacology: it appears with drugs acting on unrelated targets and after bariatric surgery, which argues against any receptor-level account as the primary cause. The finding is consistent across programmes, in harmonised IPD and in observational cohorts, and it is graded with glycaemic status. Certainty that the *association* exists is high. Certainty that it reflects *effect modification by diabetes itself*, rather than co-varying population traits, is moderate at best, because no adjusted cross-programme IPD analysis exists.

Mechanistic question. No mechanism has been measured to explain the gap, and the honest position is that three host-level hypotheses remain live and are not mutually exclusive (Section 11.1): the energetics of glycaemic correction (H1); a generalised deficit in anorectic responsiveness in the diabetic brain (H2); and degradation of the shared multi-axis gut–brain afferent pathway through which satiation signals of every kind must pass (H3).

The energetic account is the **best characterised**, which should not be mistaken for the best supported. It is quantitatively capable, operates only in the arm whose glycaemia improves, scales with the magnitude of glycaemic correction, and a formal model partially reproduces the gap from glycosuric and expenditure terms alone while predicting that stable background SGLT2 inhibition should produce the greatest attenuation of all[41]. But its mediation has never been measured; it cannot explain the deficit already present in prediabetes, where glycosuria is negligible; and the model supporting it held appetite parameters fixed at their non-diabetic values, so it excluded the main competing hypothesis by construction rather than testing against it.

The two appetite-and-signalling hypotheses predict everything the energetic account predicts, plus the prediabetes step that it cannot. Both have direct human mechanistic support — hypothalamic inhibition after oral glucose fails in BMI-matched people with T2D[70], gliosis is present in impaired glucose tolerance as well as diabetes[71], and exogenous GLP-1 loses its effect on food intake after human vagotomy[94]. Neither has been tested as a mediator of the weight gap. They rank below H1 on how well they are characterised, not on how likely they are.

The generality of the phenotype across drug classes and after bariatric surgery establishes that the cause lies in the host rather than in incretin receptor pharmacology. It does not adjudicate among the three, because all three predict it. Within the peripheral account, the narrow incretin-specific version is weak — incretin secretion is on average preserved in T2D, the portal afferent receptor appears not to be engaged by exogenous agonists, and raising native GLP-1 with DPP-4 inhibition produces no weight loss — but the broad multi-axis version is untouched by those objections. Weight-promoting co-therapies and differences in population composition are plausible additional contributors. Impaired GIP signalling, a primary endogenous secretory deficit, and intrinsic insulin-resistance-driven pharmacological resistance remain the least supported explanations.

Two observations currently discriminate, and they point in opposite directions. The deficit visible in prediabetes favours H2 and H3; the persistence of the gap under enforced, fixed energy intake[38] favours H1, though that study carries a baseline composition confounder and an opposing meta-analytic null[40]. That tension, not a settled answer, is the current state of the field.

The single most important missing piece of evidence is a matched energy-balance study that measures energy intake, total energy expenditure and urinary energy losses in people with and without T2D receiving the same incretin agent, and formally tests which components mediate the difference. Of these, **energy intake is the discriminating measurement**: equal placebo-adjusted intake suppression with a persisting weight gap would establish H1; smaller suppression in T2D would show that energetics alone cannot be the answer. The modelling gives that study sharper hypotheses — quantitative predictions for urinary glucose excretion, and a falsifiable prediction about background SGLT2 inhibition that existing trials could address by stratified reanalysis alone. Refitting the model's appetite parameters in T2D, rather than holding them fixed, would be an immediate and inexpensive first test.

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Declarations

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Methods note This is a narrative review informed by structured evidence retrieval (Supplementary Methods); it is not a systematic review and does not claim PRISMA compliance.

Supplementary information

Supplementary Methods: evidence retrieval and verification

Type: structured, targeted retrieval for a narrative review. No PRISMA screening, no duplicate independent screening, no risk-of-bias tool; not exhaustive.

Sources and tools

- PubMed/MEDLINE via PubMed MCP connector (search, metadata, PMC full text, citation matching).
- ClinicalTrials.gov via MCP connector (trial registration/eligibility; posted results not returned by the tool).
- Unpaywall API, Europe PMC, Crossref, NCBI E-utilities (enabled mid-project by network allowlisting).
- Open-access repository copies (UCL Discovery, Ulster Pure). Publisher sites (nejm.org, thelancet.com, Wiley) returned 403 to automated access.

Strategy

1. **Known-item retrieval** of landmark trials and physiology papers by citation (journal/volume/page/author), then verification of every record's abstract.
2. **Targeted searches** (examples):
    - `orforglipron[Title] AND (N Engl J Med[Journal] OR Lancet[Journal])`
    - `cagrilintide semaglutide REDEFINE`
    - `STEP-HFpEF pooled analysis semaglutide diabetes status weight`
    - `semaglutide prediabetes normoglycemia STEP trials glycemic status`
    - `tirzepatide prediabetes obesity 176 weeks diabetes prevention SURMOUNT-1`
    - `maridebart cafraglutide obesity phase 2 type 2 diabetes`
    - `(GLP-1 receptor agonist OR liraglutide OR semaglutide OR exenatide) AND energy expenditure AND (indirect calorimetry OR respiratory chamber) AND type 2 diabetes AND randomized`
    - `(GLP-1 receptor agonist ...) AND gastric emptying AND baseline gastric emptying AND type 2 diabetes AND scintigraphy`
    - `tirzepatide[Title] AND (energy intake OR appetite OR food intake) AND obesity`; `Martin CK[Author] AND tirzepatide`
    - Title-restricted searches for "(diabetes) AND (attenuated|less|lower) AND (weight loss)" — returned no dedicated comparative meta-analysis.
3. **Backward/forward tracking** from pivotal trials and pooled analyses (e.g., STEP-HFpEF pool, Perreault 2022).
4. **Peripheral incretin signalling (second round, 15 October 2026)**, prompted by the author-supplied document `Peripheral Incretin Signalling in T2DM.docx`. All 21 unique references in that document were resolved by citation matching (journal/volume/first page/author) and every record's abstract was then verified against the claim it was used for. Additional targeted searches extended the axis beyond the document:
    - `Calanna secretion of glucagon-like peptide-1 type 2 diabetes systematic review meta-analysis`; same for GIP
    - `vagotomy AND GLP-1 AND food intake` (yielded the human truncal vagotomy study and the non-vagal counter-evidence)
    - `GLP-1 receptor expression nodose ganglion diabetes obesity rodent`
    - `plasma dipeptidyl peptidase-4 activity increased type 2 diabetes compared controls`
    - `cardiovascular autonomic neuropathy prevalence type 2 diabetes review`
    - `diabetic gastroparesis prevalence scintigraphy type 2 diabetes cohort`
    - `semaglutide pharmacokinetics renal impairment hepatic impairment`
    - `(bile acid OR TGR5 OR GPBAR1) AND (GLP-1 secretion) AND (type 2 diabetes) AND human`
    - `(short-chain fatty acid OR microbiota) AND (GLP-1 secretion OR enteroendocrine) AND type 2 diabetes`
    - `(L-cell density OR enteroendocrine cell number) AND (type 2 diabetes OR obesity) AND human intestinal biopsy`
    - `preoperative diabetes predicts less weight loss after bariatric surgery`
    - `(semaglutide OR tirzepatide OR liraglutide) AND (antibodies OR immunogenicity) AND (efficacy OR exposure)`

Inclusion priorities

Matched-programme RCTs (same molecule/dose/duration/estimand) > harmonised/IPD or single-protocol cohorts > human physiology (clamps, tracers, calorimetry, ad libitum intake, fMRI) > observational cohorts > animal/cellular mechanism.

Peripheral-axis lines searched that yielded nothing usable

These were searched and are reported for completeness; none produced a source that met the inclusion priorities for a claim about diabetes status and weight response to an incretin agonist. - **Bile acid / TGR5 signalling** and **microbiome / short-chain fatty acids** as determinants of L-cell secretion in T2D: the retrieved literature concerns metformin's gut pharmacology and mechanism reviews rather than any comparison relevant to agonist response. Only the metformin-background point was retained. - **Enteroendocrine / L-cell density in human T2D intestinal biopsies**: no human study comparing L-cell density by diabetes status with a weight or agonist-response endpoint was retrieved. The reduced ileal L-cell number used in Section 5.7 is from a rodent diet-induced obesity model. - **Bariatric surgery as a parallel**: direct searches did not retrieve a usable matched comparison. The reference trail of Hall 2026 subsequently supplied one [COURCOULAS2015], which is now cited in Section 2.5 for the direction of the association. This is a retrieval failure of the original search, recorded here rather than silently repaired. - **Immunogenicity / anti-drug antibodies** as an exposure explanation: nothing retrieved that stratified efficacy by diabetes status. - **Heart rate variability or autonomic function as a predictor of GLP-1 receptor agonist weight response**: no studies retrieved. This absence is the basis for proposed study 7 in Table 4.

Third round: Hall 2026 and its evidence trail (16 October 2026)

Prompted by a reader query about a preprint published after the first draft. Retrieval: - Web search identified `Why Do Patients with Type 2 Diabetes Lose Less Weight on GLP-1s? A Mathematical Modeling Analysis`, bioRxiv 10.64898/2026.09.08.750112. - Metadata confirmed via the bioRxiv API (`api.biorxiv.org/details/biorxiv/...`) and Europe PMC (PPR1316883). Full text retrieved from the posted preprint HTML after repeated HTTP 429 rate limiting. - Its reference list was then mined and each relevant source resolved by citation matching and verified against its PubMed abstract. This supplied: the STEP UP / STEP UP T2D matched pair, the non-incretin generality evidence (naltrexone-bupropion, lorcaserin, phentermine-topiramate, bariatric surgery, formula diets), the within-T2D HbA1c gradient, and the model's own inputs (ADAG, renal threshold, the energy-expenditure coefficient). - This round materially changed the review's conclusions and corrected a sign error in proposed study 6 (see `quality_control.md`).

Note on preprint retrieval: bioRxiv rate-limits automated access aggressively. Metadata is reliably available through its API; full text required repeated retries. Preprints are not indexed in PubMed, so the PubMed-only strategy declared above would not have found this paper. That is a structural gap in the search, now explicit.

Known retrieval gaps

- Embase, Cochrane CENTRAL, conference abstracts not searched.
- **Preprint servers (bioRxiv, medRxiv) were not searched systematically.** Hall 2026 was found only because a reader asked about it. Any other relevant preprint remains unretrieved.
- Supplementary appendices of most NEJM/Lancet trials not accessible (subgroups by baseline HbA1c, background therapy, sex).
- No formal search for unpublished IPD analyses or regulatory (FDA/EMA) review documents.

Supplementary Table 1 | Structured extraction of clinical trials and cohorts

trial	agent	dose	weeks	diabetes status	n randomised	baseline bmi	baseline hba1c pct	female pct	background therapy	estimand	drug pct change	placebo pct change	etd pp	verification level	pmid
STEP 1	semaglutide s.c.	2.4 mg weekly	68	none (HbA1c<6.5% required)	1961	37.9	5.7	74.1	none	treatment policy	-14.9	-2.4	-12.4	full text (repository copy)	33567185
STEP 2	semaglutide s.c.	2.4 mg weekly (vs 1.0 mg)	68	T2D (HbA1c 7-10%)	1210	>=27 (eligibility)	7-10 (eligibility)	NR	0-3 oral agents; stratified by background therapy	treatment policy	-9.6	-3.4	-6.2	abstract	33667417
SCALE Obesity and Prediabetes	liraglutide	3.0 mg daily	56	none (61.2% prediabetes)	3731	38.3	NR	78.5	none	LOCF	-8.4 kg	-2.8 kg	-5.6 kg	abstract	26132939
SCALE Diabetes	liraglutide	3.0 mg daily (vs 1.8)	56	T2D (HbA1c 7-10%)	846	>=27	7-10	NR	0-3 OADs (metformin/TZD/SU)	NR in abstract	-6.0	-2.0	-4.0	abstract	26284720
SCALE Insulin	liraglutide	3.0 mg daily	56	T2D	396	NR	NR	NR	basal insulin + <=2 OADs; IBT	NR in abstract	-5.8	-1.5	-4.3	abstract	32139381
SURMOUNT-1	tirzepatide	5/10/15 mg weekly	72	none	2539	38.0	NR	NR	none	treatment regimen	-15.0/-19.5/-20.9	-3.1	-11.9/-16.4/-17.8 (calc)	abstract	35658024
SURMOUNT-2	tirzepatide	10/15 mg weekly	72	T2D (HbA1c 7-10%; mean 8.02; duration 8.5 y)	938	36.1	8.02	50.7	metformin 88.7%; SU 26.7%; SGLT2i 20.4%; TZD 3.5%	treatment regimen	-12.8/-14.7	-3.2	-9.6/-11.6	full text incl. supplement (repository copy)	37385275
STEP-HFpEF	semaglutide s.c.	2.4 mg weekly	52	none	529	>=30	NR	NR	HF therapy	treatment policy	-13.3	-2.6	-10.7	abstract + pooled full text	37622681
STEP-HFpEF DM	semaglutide s.c.	2.4 mg weekly	52	T2D (HbA1c<=10%)	616	>=30	NR	NR	HF therapy + glucose-lowering drugs (SU/insulin guidance)	treatment policy	-9.8	-3.4	-6.4	abstract + pooled full text	38587233
REDEFINE 1	cagrilintide+semaglutide	2.4/2.4 mg weekly	68	none	3417	NR	NR	NR	none	treatment policy	-20.4	-3.0	-17.3	abstract	40544433
REDEFINE 2	cagrilintide+semaglutide	2.4/2.4 mg weekly	68	T2D (HbA1c 7-10%)	1206	>=27	7-10	NR	NR in abstract	treatment policy	-13.7	-3.4	-10.4	abstract	40544432
ATTAIN-1	orforglipron (oral)	6/12/36 mg daily	72	none	3127	NR	NR	NR	none	treatment regimen	-7.5/-8.4/-11.2	-2.1	-5.4/-6.3/-9.1 (calc)	abstract	40960239
ATTAIN-2	orforglipron (oral)	6/12/36 mg daily	72	T2D (HbA1c 7-10; mean 8.05)	1613	35.6	8.05	46.9	NR in abstract	treatment regimen	-5.1/-7.0/-9.6	-2.5	-2.7/-4.5/-7.1	abstract	41275875
MariTide phase 2	maridebart cafraglutide	140-420 mg q4w/q8w	52	both (single protocol)	592 (465 + 127)	37.9 / 36.5	NR	63 / 42	NR	treatment policy	-12.3 to -16.2 / -8.4 to -12.3	-2.5 / -1.7	calc	abstract	40549887
retatrutide phase 2 (obesity)	retatrutide	1-12 mg weekly	48	none	338	NR	NR	48.2	none	NR	-24.2 (12 mg)	-2.1	-22.1 (calc)	abstract	37366315
retatrutide phase 2 (T2D)	retatrutide	0.5-12 mg weekly	36	T2D (HbA1c 7-10.5)	281	25-50	NR	56	diet +/- metformin	NR	-16.94 (12 mg)	-3.00	-13.9 (calc)	abstract	37385280
OASIS 1	oral semaglutide	50 mg daily	68	none	667	NR	NR	NR	none	treatment policy	-15.1	-2.4	-12.7	abstract	37385278
STEP 1/3/4 prediabetes post hoc	semaglutide s.c.	2.4 mg weekly	68	prediabetes	STEP1 subgroup 856	38.4	5.9	68.6 (sema)	lifestyle	treatment policy	-13.7 (STEP 1)	-2.4	-11.31	full text (PMC)	35724304
SURMOUNT-1 3-year prediabetes	tirzepatide	5/10/15 mg weekly	176	prediabetes	1032	NR	NR	NR	none	NR	-12.3/-18.7/-19.7	-1.3	calc	abstract	39536238
real-world cohort	semaglutide	1.7-2.4 mg weekly	26	mixed	175	41.3	NR	75.4	clinical	observed	-7.2 (T2D) vs -11.8 (no T2D) at 6 mo	none	NA	abstract	36121652
EHR cohort	tirzepatide vs semaglutide (T2D-labelled)	various	up to 52	mixed (52% T2D)	18386	39.1	NR	70.5	clinical	on-treatment	see paper (figure)	none	NA	full text (PMC)	38976257
behavioural programme	lifestyle	NA	NR	paired	24	NR	NR	50	NA	observed	-7.5 kg (T2D) vs -13.4 kg (spouses)	none	NA	abstract	3677974

Supplementary Table 2 | Mechanistic studies

source key	mechanism domain	population	model	design	measured outcome	key finding	relevance to T2D question	causal status
BLUNDELL2017	energy intake	obesity, no T2D (n=30)	human	RCT crossover 12 wk sema 1.0	ad lib intake, appetite, RMR	intake -24%; RMR unchanged	indirect (non-T2D)	causal for drug effect on intake
FRIEDRICHSEN2021	energy intake / GE	obesity, no T2D (n=72)	human	RCT 20 wk sema 2.4	ad lib intake; paracetamol GE	intake -35%; no GE delay at wk 20	indirect	causal (intake)
MARTIN2025	energy intake / reward	overweight/obesity, no T2D (n=114)	human	RCT 6 wk tirz vs lira vs pbo	ad lib intake; fMRI	-524.6 kcal; reduced food-cue activation	indirect	causal (intake)
GUTZWILLER1999	energy intake	T2D (n=12)	human	crossover IV GLP-1	ad lib intake	-27%	direct (T2D effect present)	causal
FLINT2013	energy intake	T2D (n=18)	human	crossover lira 1.8, 3 wk	ad lib intake	-18%	direct (T2D)	causal
HEISE2023	energy intake	T2D (n=117)	human	RCT 28 wk tirz/sema/pbo	appetite, ad lib intake, DXA	both reduce appetite; intake similar tirz vs sema despite more weight loss	direct (T2D)	partial
VANBLOEMENDAAL2014	CNS reward	obese T2D, obese NGT, lean (n=48)	human	crossover exenatide ± exendin 9-39, fMRI	food-cue BOLD, intake	reductions in T2D and obese; receptor-dependent	direct (both groups; magnitude not compared in abstract)	causal (receptor)
TENKULVE2015	CNS reward	T2D (n=20)	human	crossover lira vs glargine 12 wk	fMRI	effect at 10 d, not 12 wk	direct (T2D)	associational over time
FARR2016	CNS	T2D; human brain tissue	human	crossover lira; immunohistochemistry	GLP-1R localisation; fMRI	GLP-1R in hypothalamus, medulla, parietal cortex	direct (T2D)	descriptive
ROSENBAUM2022	CNS gliosis	obesity NGT/IGT/T2D (n=67)	human	cross-sectional + prospective MRI	MBH T2 relaxation	gliosis greater in IGT/T2D	direct (association)	no
SECHER2014	CNS circuits	rats/mice	animal	pharmacology, KO, electrophysiology	weight; POMC/NPY activity	ARC POMC/CART mediate lira weight loss	mechanistic only	causal in rodents

source key	mechanism domain	population	model	design	measured outcome	key finding	relevance to T2D question	causal status
GABERY2020	CNS circuits	rodents	animal	imaging, c-Fos, transcriptomics	brain access, intake, EE	CVO access; intake down without EE fall	mechanistic only	causal in rodents
KANOSKI2011	vagal/CNS	rats	animal	antagonist, deafferentation	intake	vagal + CNS GLP-1R contribute	mechanistic only	causal in rodents
BRIERLEY2021	CNS	mice	animal	chemogenetics	intake	PPG neurons not required for GLP-1RA anorexia	mechanistic only	causal in rodents
NAUCK2011	gastric emptying	healthy (n=9)	human	IV GLP-1 8.5 h	GE	rapid tachyphylaxis	indirect	causal
UMAPATHYSIVAM2014	gastric emptying	healthy men (n=10)	human	crossover infusion regimens	scintigraphic GE	prolonged exposure attenuates slowing	indirect	causal
HALAWI2017	gastric emptying	obesity (n=40)	human	RCT lira 3.0 16 wk	GE, satiation, weight	GE delay correlates with weight loss	indirect	associational
URVA2020	gastric emptying	mice; humans with/without T2D	human+animal	phase 1, 4 wk	acetaminophen GE	tachyphylaxis; residual delay in T2D	direct (T2D)	descriptive
FRASER1990	gastric emptying	T1D (n=10)	human	clamp crossover	scintigraphic GE	hyperglycaemia slows GE	indirect (T1D)	causal (glycaemia->GE)
RAYNER2020	gastric emptying	metformin-treated T2D (n=30)	human	RCT lixisenatide 8 wk	scintigraphy, tracers	sustained slowing; effect depends on baseline GE	direct (T2D)	causal
NAUCK1993	GIP/GLP-1 islet	mild T2D vs controls	human	hyperglycaemic clamp infusions	C-peptide, glucagon	GIP effect -54%; GLP-1 preserved	direct	causal (islet)
HOJBERG2009	GIP/GLP-1 islet	poorly controlled T2D (n=8)	human	4 wk insulin near-normoglycaemia	clamp C-peptide	3-4x improved incretin response	direct	causal (glucotoxicity)
XU2007	receptor expression	rat islets	animal	Px, clamp, culture	GLP-1R/GIPR mRNA	hyperglycaemia downregulates receptors	mechanistic only	causal in rodents
BERGMANN2019	GIP appetite	overweight men, no T2D (n=17)	human	crossover infusions	ad lib intake	GIP adds nothing to GLP-1 intake effect	indirect	causal (null)
ZHANG2021	GIP CNS	mice	animal	CNS-Gipr KO	intake, weight	CNS GIPR needed for co-agonist superiority	mechanistic only	causal in rodents
ADRIAENSSENS2019	GIP CNS	mice (+ human RNAscope)	animal	chemogenetics	intake	hypothalamic Gipr cells suppress intake	mechanistic only	causal in rodents
BORNER2021	GIP emesis	mice, rats, shrews	animal	pharmacology	emesis, intake	GIPR agonism blocks GLP-1RA emesis	mechanistic only	causal in animals
KILLION2018	GIPR antagonism	mice, NHP	animal	antibody	weight	antagonism lowers weight; additive with GLP-1RA	mechanistic only	causal in animals
WILLARD2020	receptor pharmacology	cells/islets	in vitro	signalling assays	cAMP, arrestin, occupancy	tirz imbalanced (GIPR) and biased	mechanistic only	pharmacology
ASMAR2010	adipose	lean healthy (n=8)	human	AV difference + clamp	adipose flux	GIP + insulin increases TAG deposition	indirect	causal (acute)
HEISE2022	insulin sensitivity	T2D (n=117)	human	RCT clamp 28 wk	M value, disposition index	tirz improves secretion, IS, glucagon	direct (T2D)	causal (drug)
ARMSTRONG2016	insulin sensitivity	NASH	human	RCT lira 12 wk tracers/clamps	hepatic/adipose IS, DNL	improved	indirect	causal (drug)
KELLEY2000	substrate	IR humans/rodents	review	synthesis	fuel selection	metabolic inflexibility	background	n/a
HALL2015	substrate vs energy	obesity (n=19)	human	metabolic ward isocaloric	fat balance by chamber	lower insulin did not increase fat loss	indirect (principle)	causal
BOGARDUS1986	glycosuria/REE	obese Pima T2D vs NGT	human	cross-sectional + 6 wk tolazamide	RMR, urine glucose, intake	RMR +5% in T2D; UGE 128->11 g/d; RMR falls	direct (T2D energetics)	causal (glycaemia->REE, UGE)
CARLSON1993	glycosuria/EE	T1D (n=6)	human	2 mo intensive insulin, constant intake, chamber	EE, glycosuria, body comp	glycosuria 428→39 mmol/d; EE -5%; +2.6 kg fat	indirect (T1D)	causal
MAKIMATTILA1999	glycosuria/REE	insulin-treated T2D (n=26)	human	RCT 12 mo insulin ± metformin	BMR, intake, glucosuria	glycaemia independent determinant of lower BMR	direct (T2D)	associational/causal
RAVE2006	renal glucose	T2D (n=22)	human	stepped clamps	UGE vs glucose	0.06→0.77 mmol/min; threshold 7.7-12.2 mM	direct (T2D)	dose-response
FERRANNINI_G2015	glycosuria energetics	T2D (n=86)	human	90 wk empagliflozin + model	weight vs glycosuric energy	206 kcal/d; intake +13% compensation	direct (T2D energetics)	model-based
POLIDORI2016	appetite feedback	T2D (n=153)	human	52 wk canagliflozin + model	calculated intake	+100 kcal/d per kg lost	direct (T2D)	model-based
LOOK2025	body composition	obesity, no T2D (n=160)	human	DXA substudy SURMOUNT-1	fat/lean	75% fat / 25% lean	indirect	descriptive
MCCRIMMON2020	body composition	T2D (n=178)	human	DXA substudy SUSTAIN 8 (no placebo)	fat/lean	fat -3.4 kg, lean -2.3 kg	direct (T2D)	descriptive
RAVUSSIN2025	EE/adaptation	obesity (no T2D) + mice	human+animal	phase 1 RCT calorimetry	EE, adaptation, RER	no adaptation change; fat oxidation up	indirect	causal (drug)
VANEYK2019	REE	T2D (n=50)	human	RCT lira 26 wk	REE, BAT	REE reduced wk 4-12	direct (T2D)	causal (drug)
HARDER2004	24h EE	obese T2D (n=33)	human	RCT lira 0.6 8 wk chamber	24h EE, DXA	no EE change	direct (T2D)	causal (null)
GOLUBIC2024	energy balance	T2D (n=28 randomised)	human	RCT cotadutide 42 d DLW + IC	EI, EE	weight loss from EI; small EE change	direct (T2D)	causal (drug)
UKPDS33	co-therapy	newly diagnosed T2D (n=3867)	human	RCT	weight	insulin +4.0 kg; SU +1.7-2.6 kg	direct (T2D)	causal (drug)
WING1987	behaviour	T2D + nondiabetic spouses (12 pairs)	human	paired behavioural programme	weight, intake	7.5 vs 13.4 kg; smaller intake reduction in T2D	direct comparison	associational
CALANNA_GLP1	peripheral: endogenous secretion	T2D n=275 vs non-T2D n=279 (22 trials)	human	meta-analysis of OGTT/meal tests	total plasma GLP-1 (peak/tAUC/iAUC)	no significant difference overall; HbA1c and FPG predicted iAUC	direct: contradicts uniform secretory deficit; supports graded effect	associational
CALANNA_GIP	peripheral: endogenous secretion	T2D n=363 vs controls n=325 (23 trials)	human	meta-analysis of OGTT/meal tests	total plasma GIP	secretion preserved; iAUC lower (WMD -648 pmol/L.min); age/HbA1c negative predictors	direct	associational
TOFTNIELSEN2001	peripheral: endogenous secretion	54 T2D vs 33 NGT vs 15 IGT	human	4-h mixed meal	GLP-1 AUC 0-240 min	2482+/-145 vs 3101+/-198 pmol/L.240min (p=0.024)	direct	associational
VILSBOLL2003	peripheral: endogenous secretion	8 obese T2D vs 8 obese healthy	human	two meal sizes (260/520 kcal)	GLP-1 and GIP responses	GLP-1 reduced in T2D; GIP normal	direct	associational
ZHENG2015	peripheral: degradation	985 newly diagnosed T2D	human	cross-sectional	plasma DPP-4 activity	positively associated with HbA1c and HOMA-IR	indirect; no non-diabetic comparator	associational
MALIN2013	peripheral: degradation	14 obese adults with metabolic syndrome	human	12-wk exercise intervention	plasma DPP-4; clamp insulin sensitivity	DPP-4 -10%; change correlated with insulin sensitivity (r=-0.59)	weak; n=14, no comparator, GLP-1 and weight not measured	associational
YAMASHITA2019	peripheral: degradation	mice (normal and T2D model)	animal	oral vs subcutaneous anagliptin	small-intestinal vs plasma DPP-4 inhibition; glucose	glucose lowering tracked intestinal (r=0.949) not plasma DPP-4 inhibition	mechanistic: locates degradation in gut	causal in model
WILLARD2017	peripheral: degradation	neprilysin-deficient vs WT mice	animal	16 wk 60% fat diet	active GLP-1; plasma DPP-4; OGTT	active GLP-1 5.8 vs 3.5 pmol/L; lower DPP-4; better tolerance	indirect; genetic deletion not pharmacology	causal in model

source key	mechanism domain	population	model	design	measured outcome	key finding	relevance to T2D question	causal status
NISHIZAWA2000	peripheral: portal sensing	anaesthetised rats	animal	intraportal injection + hepatic vagal afferent recording	afferent discharge rate	GLP-1(7-36) facilitated afferents; exendin-4 did not at any dose; exendin(9-39) did not block	ARGUES AGAINST agonist engagement of portal route	causal in model
EDGERTON2013	peripheral: portal sensing	conscious dogs	animal	hyperglycaemic clamp + intraportal exenatide	hepatic glucose production/uptake	no insulin-independent hepatic effect of exenatide	argues against direct hepatic incretin effect	causal in model
HAYES2011	peripheral: vagal route	rats	animal	common hepatic branch ablation vs subdiaphragmatic deafferentation	OGTT; intake suppression by GLP-1	hepatic branch not required; full deafferentation blunted both	refines route: paracrine GI afferents not hepatoportal	causal in model
ZHANGRITTER2011	peripheral: vagal route	rats	animal	vagotomy / capsaicin + IV GLP-1	sucrose intake; hindbrain c-Fos	intake suppression not diminished (enhanced at high dose); c-Fos comparable	COUNTER-EVIDENCE: circulating GLP-1 acts non-vagally	causal in model
IWASAKI2018	peripheral: vagal route	mice and rats	animal	oral D-allulose +/- vagotomy / GLP-1R blockade / afferent-selective KO	GLP-1 release; intake; glucose tolerance	effects blunted by vagotomy and vagal-afferent GLP-1R inactivation	supports vagal route for locally released GLP-1	causal in model
KNAUF2008	peripheral: portal sensing	mice incl. Glp1r-KO and high-fat-fed	animal	low-rate intragastric glucose	muscle glycogen synthesis; brain c-Fos	+30% glycogen synthesis; abolished in Glp1r-KO and by central exendin(9-39); prevented by high-fat feeding	central not portal receptor dependence; HFD abolishes signal	causal in model
PLAMBOECK2013	peripheral: vagal route	20 men truncally vagotomised + pyloroplasty vs 10 matched controls	human	GLP-1 vs saline infusion; meal + ad libitum meal	ad libitum energy intake; gastric emptying; insulin	intake reduced in controls, NO effect in vagotomised; emptying slowed only in controls	STRONGEST human evidence vagal route is needed for native GLP-1; not an agonist, no weight endpoint	causal in humans (native GLP-1)
DUCA2013	peripheral: agonist response	obese-prone vs obese-resistant rats	animal	intraperitoneal exendin-4 on chow vs high-fat diet	food intake; nodose GLP-1R mRNA; L-cell number	obese-prone suppressed less at all doses on HF diet, with nodose GLP-1R downregulation and reduced ileal L-cells	CLOSEST analogue of clinical observation: agonist effect blunted in dysmetabolic host	causal in model
LEE2012	peripheral: enteric/vagal sensing	UCD-T2DM rats (prediabetic/recent/3-month diabetic)	animal	orogastric glucose gavage; pCaMKII immunohistochemistry	activation of enteroendocrine cells / enteric and vagal afferent neurons	activation markedly reduced with diabetes duration; GLP-1 immunoreactivity reduced (GIP not)	direct in a T2D model	causal in model
MALBERT2021	peripheral: portal sensing	insulin-resistant obese animals	animal	68Ga GLP-1R PET + single-neuron electrophysiology	portal GLP-1R binding potential; afferent firing	afferents failed to modulate firing; reduced binding potential between splenic vein and hepatic inlet	direct in insulin resistance	causal in model
BONORA2020	peripheral: autonomic neuropathy	806 newly diagnosed T2D	human	cross-sectional systematic assessment	cardiovascular autonomic reflex tests	CAN 18.6%; somatic neuropathy 21.2%	prevalence only; proxy for vagal integrity	descriptive
DUCHEN1980	peripheral: autonomic neuropathy	5 post-mortem cases, TYPE 1 diabetes	human	post-mortem histopathology	vagus and sympathetic trunk fibre counts	severe loss of myelinated vagal fibres; ganglion and inflammatory changes	type 1, n=5, established autonomic failure; limited transfer to T2D	descriptive
ROBERTSON1980	peripheral: autonomic neuropathy	db/db mice	animal	morphometry + electrophysiology	nerve fibre counts; conduction velocity	axonal loss and shrinkage; vagal changes appeared late	indirect	causal in model
CHEDID2019	gastrointestinal	108 referred patients with diabetes (71.3% T2D)	human	scintigraphic GE + SPECT accommodation	gastric emptying rate	abnormal in 56%: RAPID 37%, slow 19%; 40.3% of T2D accelerated at 1 h	direction of GE change in T2D is heterogeneous and often accelerated	descriptive
OVERGAARD2021PK	drug exposure	healthy, renal/hepatic impairment, and T2D	human	population PK across 6 trials	oral semaglutide bioavailability	no significant difference healthy vs T2D; within-subject variability 137% → 33% at steady state	argues against reduced exposure in T2D	associational
MCCREIGHT2016	clinical: background therapy	—	human	review	metformin gut effects	raises GLP-1 and intestinal bile acid pool; alters microbiome	background metformin modifies endogenous axis opposite to deficit hypothesis	review
HOLT2019	energy intake (central)	mice	animal	viral ablation / chemogenetics of NTS PPG neurons	ad libitum and post-fast intake; body weight	no change in basal intake or weight; increased post-fast intake; stress hypophagia abolished	secondary satiation circuit not primary regulator	causal in model
HALL2026	energy losses + expenditure	simulated STEP UP (no T2D) and STEP UP T2D	model	energy-balance model; appetite parameters fit to non-T2D then held fixed	UGE from eGFR/renal threshold/HbA1c; EE from fasting glucose	partially reproduces the T2D gap from energetic terms alone; stable background SGLT2i predicted greatest attenuation	most direct quantitative treatment of the question; simulation not measurement	modelled mediation
STEPUP	clinical phenotype	obesity without T2D, n=1407	human	RCT 72 wk, semaglutide 7.2 vs 2.4 vs placebo	% weight change	-18.7% (7.2 mg) vs -3.9% placebo; ETD -14.8	non-T2D arm of the cleanest matched pair	causal (drug effect)
STEPUP_T2D	clinical phenotype	obesity with T2D, n=512, HbA1c 8.1%	human	RCT 72 wk, semaglutide 7.2 vs 2.4 vs placebo	% weight change	-13.2% vs -3.9% placebo; ETD -9.3; ratio 0.63	T2D arm; identical placebo change to STEP UP; ~30% on background SGLT2i	causal (drug effect)
GOUGEON2002	energy expenditure	65 obese adults with T2D	human	cross-sectional indirect calorimetry	resting energy expenditure	REE elevated up to 8% with poor glycaemic control; FPG improves prediction	source of the EE coefficient used in HALL2026	associational
HOLLANDER2013	clinical: non-incretin	overweight/obese with T2D, n=505	human	RCT 56 wk naltrexone-bupropion	% weight change	-5.0% vs placebo -1.8%	T2D gap with a non-incretin drug	causal (drug effect)
ONEIL2012	clinical: non-incretin	T2D, n=604	human	RCT 1 yr lorcaserin	% weight change	-4.5% to -5.0% vs placebo -1.5%	T2D gap with a 5-HT2C agonist	causal (drug effect)
HENRY1986	energy balance under fixed intake	10 obese T2D vs 5 obese non-diabetic	human	36-day metabolic ward at 300 kcal/day	weight loss rate; glycaemia	308 vs 450 g/day (p<0.01); FPG 291→95 mg/dL in T2D	gap persists at enforced equal intake; but baseline weight/LBM differed	causal with confounder
LESLIE2017	energy balance under fixed intake	n=569 across 5 studies	human	meta-analysis of formula diets	final weight loss	no significant difference by diabetes status (-1.2 kg; -4.1 to 1.6)	null; counterweight to HENRY1986	associational
COURCOULAS2015	clinical: non-pharmacological	bariatric surgery cohort (LABS)	human	prospective cohort 3 yr	weight change	preoperative diabetes predicts smaller weight loss	gap after surgery, no drug involved	associational

Supplementary Table 3 | Claim-to-evidence ledger

claim id	claim	source keys	source access	verification status	caveat
C01	Placebo-adjusted weight loss with sema 2.4 is -12.4 pp without diabetes vs -6.2 pp with T2D	STEP1; STEP2	abstract + STEP1 full text	verified	between-trial comparison
C02	Across 8 matched comparisons T2D/no-T2D ETD ratio median 0.62 (0.50-0.78)	Table 1 sources	derived (analysis/derived.py)	verified calculation	pairs not independent; between-trial
C03	STEP-HFpEF pooled IPD heterogeneity for weight p<0.0001, I2 95.8%; ~40% less weight loss in T2D	HFpEF_POOL	PMC full text	verified	diabetes not randomised

claim id	claim	source keys	source access	verification status	caveat
C04	Placebo-arm weight change not smaller in T2D trials	STEP1/2, SURMOUNT1/2, ATTAIN1/2, REDEFINE1/2, STEP-HFpEF/DM	abstracts	verified	MariTide placebo -1.7 vs -2.5 is the exception
C05	Prediabetes subgroup in STEP 1 lost 13.7% (ETD -11.31); implied normoglycaemic ~15.9%	PERREAULT2022; STEP1	PMC full text + derived	verified (derived value approximate)	weighted-mean assumption
C06	GI AEs 72.5% (REDEFINE 2) vs 79.6% (REDEFINE 1); AE discontinuations similar ATTAIN-1/2	REDEFINE1/2; ATTAIN1/2	abstracts	verified	
C07	SURMOUNT-2 efficacy-estimand ETD -10.1/-12.4	SURMOUNT2	full text	verified	
C08	Semaglutide exposure determined chiefly by body weight (T2D popPK)	CARLSSON2018; OVERGAARD2021	abstracts	verified	T2D-only data; cross-population exposure not compared
C09	Strathe model used baseline HbA1c as covariate; effect size not verified	STRATHE2023	abstract	partially verified	full text not retrievable (Wiley 403)
C10	Incretin effect 36% (T2D) vs 73% (controls)	NAUCK1986	abstract	verified	
C11	Reduced incretin effect is a consequence of the diabetic state	KNOP2007; HOJBERG2009	abstracts	verified (as authors' interpretation)	
C12	GLP-1 insulinotropic activity preserved, GIP reduced 54% in mild T2D	NAUCK1993	abstract	verified	mild T2D, HbA1c 6.3%
C13	Appetite/intake suppressed by GLP-1 agonism in T2D	GUTZWILLER1999; FLINT2013; HEISE2023; VANBLOEMENDAAL2014	abstracts	verified	
C14	No study compares appetite suppression magnitude by diabetes status under matched conditions	search (PubMed)	search negative	not found	absence-of-evidence claim; limited by search scope
C15	Tirz vs sema in T2D: intake did not differ despite greater weight loss	HEISE2023	abstract	verified	single test meal
C16	Hypothalamic gliosis greater in IGT/T2D	ROSENBAUM2022	abstract	verified	association
C17	GE slowing subject to tachyphylaxis	NAUCK2011; UMAPATHYSIVAM2014	abstracts	verified	contradicted in part by JONES2020 (exenatide QW)
C18	GIP does not augment GLP-1 intake suppression in humans	BERGMANN2019	abstract	verified	acute infusion, no T2D
C19	Attenuation ratio similar across GIPR agonist, antagonist and GLP-1-only agents	Table 1	derived	verified calculation	between-trial
C20	Isocaloric carb restriction lowers insulin but not more fat loss	HALL2015	abstract	verified	6-day ward study
C21	No human T2D study measured muscle glucose uptake, oxidation, glycogen and total EE together during incretin therapy	search (PubMed)	search negative	not found	absence-of-evidence claim
C22	Glycogen-water shifts could change weight independent of fat	general physiology	inference	unsourced hypothesis	labelled as inference in text
C23	UKPDS: insulin +4.0 kg, SU +1.7-2.6 kg	UKPDS33	abstract	verified	DOI derived from PII
C24	Lira 3.0 ETD -4.3 in insulin-treated T2D vs -4.0 without insulin	SCALE_INSULIN; SCALE_DM	abstracts	verified	IBT in SCALE Insulin
C25	UGE 0.06→0.77 mmol/min at 7.8→13.3 mM; threshold 7.7-12.2	RAVE2006	abstract	verified	
C26	Sulfonylurea: UGE 128→11 g/d; RMR fall; intake cut to hold weight	BOGARDUS1986	abstract	verified	Pima, FPG 253 mg/dL
C27	Intensive insulin: glycosuria 428→39 mmol/d; EE -118 kcal/d; +2.6 kg fat	CARLSON1993	abstract	verified	T1D
C28	Improved glycaemia lowers BMR (independent determinant)	MAKIMATTILA1999	abstract	verified	authors' conclusion
C29	SGLT2i glycosuria 206 kcal/d; observed weight 29% of predicted; intake +13%	FERRANNINI_G2015	abstract	verified	model-based
C30	100 kJ/d → ~1 kg steady state; half in ~1 y; fat 39.5 MJ/kg, lean 7.6 MJ/kg	HALL2011	PMC full text	verified	
C31	Scenario offsets ~1 to ~6.7 kg by 68-72 wk	analysis/derived.py	calculation	assumption-driven	inputs not measured in incretin trials
C32	Tirz DXA: 75% fat, 25% lean	LOOK2025	abstract	verified	no T2D
C33	Incretin agonists do not raise EE in humans	BLUNDELL2017; HARDER2004; VANEYK2019; RAVUSSIN2025; GOLUBIC2024	abstracts	verified	REE may fall
C34	T2D lost less weight than nondiabetic spouses in behavioural programme	WING1987	abstract	verified	n=12 pairs
C35	Real-world: T2D 7.2% vs 11.8% at 6 mo (sema); no-T2D lost more on both drugs (EHR)	GHUSN2022; RODRIGUEZ2024	abstract; PMC full text	verified	observational
C36	MariTide T2D cohort -8.4 to -12.3% vs obesity -12.3 to -16.2%	MARITIDE2025	abstract	verified	dose arms differ
C37	Endogenous GLP-1 secretion is reduced in T2D in individual studies but preserved on meta-analysis; reduction is a function of glycaemic control not diabetes per se	TOFTNIELSEN2001; VILSBOLL2003; CALANNA_GLP1; CALANNA_GIP	abstract	verified	meta-analyses are the stronger evidence; the doc premise of a uniform secretory deficit is not supported
C38	Plasma DPP-4 activity correlates with insulin resistance within T2D	ZHENG2015; MALIN2013	abstract	verified with correction	NOT verified that DPP-4 is higher in metabolic syndrome or T2D than in people without; no non-diabetic comparator in either source; Malin n=14 and is an exercise intervention
C39	Neprilysin is a second degradative route for GLP-1	WILLARD2017	abstract	verified with correction	genetic neprilysin deletion in mice, NOT pharmacological DPP-4 reduction; animal
C40	Degradation matters mainly in the gut and portal bed rather than systemically	YAMASHITA2019	abstract	verified	rodent; correlation across doses, not a causal partition
C41	The hepatic vagal response to intraportal GLP-1 uses a receptor not engaged by exendin-4 and not blocked by exendin(9-39)	NISHIZAWA2000; EDGERTON2013	abstract	verified	argues AGAINST agonist engagement of the portal route; receptor never molecularly identified; anaesthetised rats, 2000; never repeated with a long-acting agonist
C42	The hepatoportal vagal branch is not required for GLP-1's incretin or intake-suppressing effects; paracrine GI vagal afferents are	HAYES2011	abstract	verified	animal; refines and partly contradicts the portal hypothesis
C43	Circulating GLP-1 can suppress food intake without vagal afferents	ZHANGRITTER2011	abstract	verified	animal; counter-evidence to the vagal hypothesis
C44	Intact vagal innervation is required for exogenous GLP-1 to reduce food intake in humans	PLAMBOECK2013	abstract	verified	20 men, surgical truncal vagotomy + pyloroplasty, native GLP-1 not an agonist, no body-weight endpoint; pyloroplasty confounds emptying
C45	Cardiovascular autonomic neuropathy is present in 18.6% of patients at T2D diagnosis	BONORA2020	abstract	verified	a cardiovascular proxy, not a measure of the gastrointestinal afferent limb
C46	Reported CAN prevalence ranges from about 2.5% to as high as 90% depending on tests and cohort	AGASHE2018	abstract	verified	review; the upper figure refers to type 1 diabetes. The doc's '20-40% in known disease' was NOT verified and is not used
C47	Vagal myelinated fibre loss is documented post mortem in diabetes	DUCHEN1980	abstract	verified with correction	TYPE 1 diabetes, n=5, established clinical autonomic failure. The doc described this as 'individuals with diabetes'; transfer to T2D trial populations is limited
C48	Diabetic mice show structural peripheral nerve changes including the vagus	ROBERTSON1980	abstract	verified with correction	db/db mice; vagal changes appeared LATE relative to other nerves
C49	Gut glucose sensing and enteric/vagal afferent activation are impaired in a type 2 diabetes model	LEE2012	abstract	verified with correction	UCD-T2DM RATS, not mice as stated in the source document; no agonist given; no weight endpoint

claim id	claim	source keys	source access	verification status	caveat
C50	Portal GLP-1 receptor density and afferent glucose responsiveness are reduced in insulin-resistant obese animals	MALBERT2021	abstract	verified with correction	reduction was between the SPLENIC VEIN and the entrance of the liver, not 'porta hepatis and left gastric vein' as stated in the source document; species not stated in the abstract
C51	Intragastric glucose signalling to the hypothalamus requires GLP-1 receptor signalling and is abolished by high-fat feeding	KNAUF2008	abstract	verified with correction	the receptor dependence demonstrated was CENTRAL (central exendin(9-39), Glp1r-KO), not portal; c-Fos colocalised with arcuate NPY neurons; animal
C52	Agonist-induced anorexia is blunted in a dysmetabolic rodent host alongside nodose GLP-1 receptor downregulation and reduced ileal L-cell number	DUCA2013	abstract	verified	the only identified study testing an agonist alongside peripheral changes; diet-induced obesity model not diabetes; short-acting agonist; acute intake endpoint
C53	Gastric emptying in diabetes referred for upper GI symptoms is more often accelerated than delayed	CHEDID2019	abstract	verified	referral population with symptoms; not population-representative
C54	Oral semaglutide bioavailability does not differ significantly between healthy participants and people with T2D	OVERGAARD2021PK	abstract	verified	population PK model; within-subject variability in bioavailability is high (137%) but falls to 33% at steady state
C55	Background metformin raises endogenous GLP-1 and alters the intestinal bile acid pool and microbiome	MCCREIGHT2016	abstract	verified	review; no quantification, and no study has tested whether background metformin modifies weight response to an incretin agonist
C56	NTS preproglucagon neurons form a secondary satiation circuit and are not required for basal intake or weight regulation	HOLT2019	abstract	verified	animal
C57	The deficit appears with agents that do not depend on the endogenous incretin axis or on DPP-4 (orforglipron, cagrilintide-semaglutide), which bounds the peripheral explanation	ATTAIN1; ATTAIN2; REDEFINE1; REDEFINE2	abstract	verified reasoning	argument from the cross-agent pattern in Table 1, not a direct test; orforglipron ratios are the highest observed (0.71-0.78), which is weakly consistent in the opposite direction
C58	No study has measured endogenous incretin secretion, DPP-4 activity or autonomic function as predictors of weight response to an incretin agonist in either population	—	absence of evidence	unverifiable (PubMed-only search)	absence-of-evidence statement limited by the search scope stated in search_strategy.md
C59	Semaglutide 7.2 mg: ETD -14.8 pp without diabetes (STEP UP) vs -9.3 pp with T2D (STEP UP T2D); ratio 0.63	STEPUP; STEPUP_T2D	abstract	verified	cleanest matched pair: same programme, dose, duration and estimand, and identical placebo-arm change (-3.9%) in both
C60	Across nine matched comparisons the T2D/no-T2D ETD ratio is median 0.63 (range 0.50-0.78); absolute gap median 5.5 pp (1.8-6.9)	Table 1 sources	derived (analysis/derived.py)	verified calculation	pairs not independent (multiple doses from the same trials); between-trial; no pooling or CIs computed
C61	The T2D weight-loss gap is not specific to incretin drugs: it appears with naltrexone-bupropion, lorcaserin, phentermine-topiramate and after bariatric surgery	HOLLANDER2013; GREENWAY2010; ONEIL2012; SMITH2010; GADDE2011; COURCOULAS2015	abstract	partially verified	T2D-arm numbers verified for HOLLANDER2013 and ONEIL2012 only; the non-diabetes comparators (GREENWAY2010, SMITH2010), GADDE2011 and COURCOULAS2015 are cited for direction and were NOT independently number-verified
C62	Under enforced 300 kcal/day on a metabolic ward, T2D subjects lost 308 g/day vs 450 g/day in non-diabetic subjects	HENRY1986	abstract	verified with caveat	the non-diabetic group had significantly greater initial body weight and lean body mass, stated by the authors as a confounder; n=10 vs 5; 36 days
C63	A meta-analysis of low-energy formula diets found NO significant weight-loss difference by diabetes status (-1.2 kg; 95% CI -4.1 to 1.6)	LESLIE2017	abstract	verified	null result; 5 studies, n=569; the counterweight to C62. HALL2026's attribution of the null to one dominant study is his interpretation, reported as such
C64	A formal energy-balance model partially reproduces the STEP UP T2D weight-loss gap using only UGE and energy-expenditure terms, with appetite parameters held fixed from the non-diabetic fit	HALL2026	full text of preprint	verified	PREPRINT, not peer reviewed; funded by AstraZeneca and authored by an AstraZeneca employee; simulation not measurement; the author states the explanation is PARTIAL and recommends direct measurement
C65	The model predicts that stable background SGLT2i therapy produces the GREATEST attenuation of semaglutide weight loss, while co-initiation of SGLT2i and GLP-1 produces greater weight loss than either alone	HALL2026	full text of preprint	verified	untested prediction; STEP UP T2D did not report outcomes by SGLT2i status so the simulations were averaged 30:70; corrects the opposite-signed rationale previously given for proposed study 6
C66	Resting energy expenditure in obese T2D is elevated by up to 8% with poor glycaemic control and is better predicted when fasting glucose is included	GOUGEON2002	abstract	verified	cross-sectional, n=65; this is the source of the 3.5 kcal/day per mg/dL coefficient used in HALL2026
C67	Weight reduction with tirzepatide varied by baseline HbA1c category in SURMOUNT-2	SATTAR2025	title only	partially verified	PubMed record carries NO abstract (Letter); cited for the direction stated in the title; no numbers taken
C68	Hall 2026 quotes STEP UP and STEP UP T2D weight losses higher than the published treatment-policy values	HALL2026; STEPUP; STEPUP_T2D	full text + abstracts	discrepancy flagged	preprint quotes ~18%/~21% and ~11%/~14%; publications report -15.6%/-18.7% and -13.2% (treatment policy). Consistent with a trial-product estimand but the preprint does not state which. This review uses the published treatment-policy values
C69	Glucose ingestion fails to inhibit hypothalamic neuronal activity in T2D, in BMI-comparable controls	VIDARSDOTTIR2007	abstract	verified	n=7 vs 10; men only; 2007; oral glucose not an agonist; no weight or intake endpoint. Direct human evidence of impaired central nutrient sensing independent of adiposity
C70	The roles of ghrelin, CCK, GLP-1 and PYY in eating in obesity and after RYGB remain uncertain	STEINERT2017	abstract	verified	review; the multi-axis satiety hypothesis is under-tested rather than refuted
C71	Bile acid, CCK and L-cell GLP-1 signalling are one interlinked system, perturbable in T2D	BRONDEN2017	abstract	verified	n=15 metformin-treated T2D; sevelamer abolished bile-acid-induced GLP-1 rise; appetite and ad libitum intake measured but values not in abstract
C72	Cross-intervention generality of the gap does not discriminate between the three host-level hypotheses	Section 2.5 sources	reasoning	corrected	CORRECTION to the previous draft, which inferred energetics from generality. All three hypotheses predict generality. See quality_control.md and Section 11.1
C73	The Hall model could not express a blunted appetite response because appetite parameters were held fixed at non-diabetic values	HALL2026	full text of preprint	verified	a blunted T2D appetite response could appear only as residual unexplained gap; the model cannot distinguish energetics-sufficient from energetics-plus-appetite

Supplementary Note: quality control, limitations and analyses not performed

Verification performed

- **Bibliography:** 141 of the 142 cited references resolve to PubMed records with DOIs. The exception is Hall 2026, a bioRxiv preprint with no PubMed record; it is handled by an explicit manual entry in `analysis/build_manuscript.py` and its full text was retrieved and read directly. The bibliography was generated programmatically from NCBI esummary. For UKPDS 33 the PubMed record has no DOI, so the DOI was derived from the publisher PII and is flagged as such.
- **Per-source verification:** for every source, the population, design and the specific numbers used in the text were checked against the abstract (all sources) or the full text. The log is in `references/verified_log.md`, and claims are mapped in `evidence/claim_evidence_ledger.csv`.
- **Full texts read:** STEP 1 and SURMOUNT-2 including its supplement (open-access repository copies), the STEP-HFpEF pooled analysis, Perreault 2022, Rodriguez 2024 and Hall 2011 (PMC).
- **Corrections made during audit:**
  - i. A cross-drug comparison of efficacy estimands (tirzepatide vs semaglutide) was removed.
  - ii. An overstatement that liraglutide 0.6 mg “normalised HbA1c” was corrected to “improved glycaemia”.
  - iii. An unsourced claim that agonists “reset” the appetite set point was relabelled as an untested interpretation.

Corrections made while integrating the peripheral-signalling source document (Section 5)

The author-supplied document `Peripheral Incretin Signalling in T2DM.docx` was verified source by source against PubMed. All 21 of its unique references resolved to PubMed records. Eleven of its claims required correction before use, and these corrections are reflected in Section 5 and logged in `evidence/claim_evidence_ledger.csv` (C37-C58):

1. **Reduced GLP-1 secretion in T2D was presented as established.** Two meta-analyses (22 and 23 trials) find incretin secretion preserved on average in T2D, with reduction predicted by HbA1c and fasting glucose rather than by diabetes status [CALANNA_GLP1, CALANNA_GIP]. Section 5.2 now presents the individual studies and the meta-analyses together, and the graded-with-glycaemia reading replaces the uniform-deficit reading.

2. **DPP-4 “hyperactivity” in metabolic syndrome.** Neither cited source establishes that DPP-4 activity is higher than in people without diabetes. Zheng 2015 is cross-sectional within newly diagnosed T2D with no non-diabetic comparator; Malin 2013 is a 12-week exercise intervention in 14 people that measured neither GLP-1 nor body weight. The claim was narrowed to a within-T2D association with insulin resistance.
3. **Willard 2017 was described as pharmacological DPP-4 reduction normalising active GLP-1.** It is a genetic neprilysin-deficiency study in mice; lower DPP-4 activity was an associated finding, not an intervention. Reframed as evidence for neprilysin as a second degradative route.
4. **Duchen 1980 was described as “individuals with diabetes”.** It is five post-mortem cases of early-onset *insulin-dependent* (type 1) diabetes with established clinical autonomic failure. Now stated explicitly, with its limited transfer to T2D populations.
5. **Lee 2012 was described as mice.** The study used UCD-T2DM *rats*, compared against prediabetic rats, not “prediabetic control mice”.
6. **Malbert 2021 anatomical site.** The reduction in GLP-1 receptor binding potential was between the *splenic vein and the entrance of the liver*, not “between the porta hepatis and left gastric vein”. Species is not stated in the abstract, so the porcine model is not asserted.
7. **Knauf 2008 receptor dependence.** The abolition of the response was demonstrated with *central* exendin 9-39 infusion and in GLP-1 receptor knockout mice; this is central, not portal, receptor dependence. The finding that high-fat feeding prevented the response entirely was not used in the source document and has been added, being more relevant.
8. **Autonomic neuropathy prevalence.** The figure of 18.6% at diagnosis is verified [BONORA2020]. The source document’s “20 to 40% of those with known disease” could not be verified from the cited source and is not used; the reported range is instead given as very wide and test-dependent [AGASHE2018].
9. **The Nishizawa/Edgerton findings cut against the document’s own thesis.** The source document states them correctly but treats them as compatible with a peripheral explanation of attenuated weight loss. Because the portal afferent receptor is not engaged by exendin-4 or exenatide, loss of endogenous portal signalling in T2D would lower the baseline rather than blunt the drug. Section 5.4 states this as the principal problem for the hypothesis.
10. **Numeric physiology claims drawn from reviews were dropped.** The document’s “50-70% of the total postprandial insulin response”, “more than 50% of an intravenous bolus inactivated within two minutes” and “half-life approximately 1-2 minutes” come from review articles whose abstracts do not contain them. Section 5.1 makes these points qualitatively, citing the reviews, and Section 3 retains the verified primary figures (36% vs 73% incretin contribution [NAUCK1986]).
11. **Hepatoportal emphasis.** The document frames the vagal route as hepatoportal. Hayes 2011 shows the common hepatic branch is not required, while complete subdiaphragmatic deafferentation is; the route is therefore described as paracrine signalling on gastrointestinal vagal afferents.

Material added beyond the source document, from new searches: human truncal vagotomy evidence [PLAMBOECK2013]; non-vagal counter-evidence [ZHANGRITTER2011]; route refinement [HAYES2011]; the only agonist-response study with measured peripheral changes [DUCA2013]; the two secretion meta-analyses; neprilysin; background metformin as a confounder acting opposite to the hypothesis [MCCREIGHT2016]; heterogeneity and frequent acceleration of gastric emptying in diabetes [CHEDID2019]; oral semaglutide bioavailability in T2D [OVERGAARD2021PK]; NTS preproglucagon circuit characterisation [HOLT2019].

Integration of Hall 2026 (bioRxiv preprint) — 16 October 2026

A modelling preprint directly addressing this review’s central question appeared after the first draft was completed and has been integrated into Section 9.3 [HALL2026]. Handling:

- **Full text retrieved and read**, not just the abstract. Metadata cross-checked against the bioRxiv API and Europe PMC (PPR1316883).
- **It is a preprint and has not been peer reviewed.** This is stated in the bibliography entry, in Section 9.3 and in Section 12.
- **Competing interests are stated in the text.** The study is funded by AstraZeneca and the author is an employee of AstraZeneca Pharmaceuticals. The analysis is not discounted for this — its assumptions are explicit and independently checkable — but readers are told.
- **It is a simulation, not a measurement.** Section 9.3, Section 12 and the conclusions all state that a model reproducing an observation does not establish that the modelled mechanism produced it. The preprint’s own conclusion is that the energetic terms “may *partially* explain” the gap, and this review does not upgrade that to a full explanation.
- **A numerical discrepancy with the primary trial reports is flagged**, in Section 9.3 and ledger row C68: the preprint quotes STEP UP weight losses of $_{18\%}$21% and STEP UP T2D of $_{11\%}$14%, where the publications report treatment-policy values of -15.6%/-18.7% and -13.2%. This is consistent with a trial-product estimand, but the preprint does not say which estimand it simulated. Table 1 uses the published treatment-policy values throughout.

Reasoning error in the previous draft, corrected (16 October 2026)

Section 2.5 argued: the gap appears with non-incretin drugs and after bariatric surgery, therefore it is not receptor-level, therefore the energetics of glycaemic correction are the explanation. **The second inference does not follow.** Cross-intervention generality rules out mechanisms specific to incretin receptor pharmacology; it does not select among host-level mechanisms, and at least three are compatible with it:

1. energetics of glycaemic correction;
2. a generalised deficit in anorectic responsiveness in the diabetic brain;
3. degradation of the shared multi-axis gut-brain afferent pathway.

All three predict cross-intervention generality, so that observation cannot discriminate. The error mattered in three places, all now corrected:

- **Section 2.5** previously concluded that generality “raises the weight of explanations rooted in the energetics of the diabetic state.” It now states what the inference does and does not license, and names the three competing classes.
- **Section 5.8** previously used generality to *bound* the peripheral hypothesis, asserting that “a defect in peripheral incretin signalling cannot explain a deficit that persists when no incretin drug is given.” That is true only of the narrow, incretin-specific version. The vagal afferent pathway described in 5.4-5.6 carries CCK, PYY, ghrelin, distension and nutrient signals, so the broad version predicts exactly the observed generality. The bound has been reversed and the mistake stated in the text.
- **Section 11 and the conclusions** previously ranked energetics as uniquely accounting for all robust features of the phenotype. They now distinguish *best characterised* from *best supported*, and note that the competing hypotheses additionally explain the prediabetes step, which energetics cannot.

Bariatric surgery was cited as evidence against a receptor-level cause. It is in fact neutral between hypotheses 1 and 3, because RYGB works substantially through large postoperative rises in GLP-1 and PYY and therefore depends on an intact receiving apparatus. This is now stated.

New Section 11.1 sets out the discriminating predictions. A further methodological point is recorded there: the Hall model held appetite parameters fixed at their non-diabetic values, so it could not express hypothesis 2 at all; a blunted appetite response could only appear as residual unexplained gap, which is what the analysis reports. The model therefore cannot distinguish “energetics explains part and nothing else is needed” from “energetics explains part and blunted anorectic response explains the rest.” Table 4 study 3b proposes the direct test (refit the appetite parameters in T2D).

Error in the previous draft, corrected

Proposed study 6 in Table 4 previously argued that co-administering an SGLT2 inhibitor would *maintain* glycosuria and therefore *preserve* weight loss in T2D, bringing it closer to non-diabetes levels. **This reasoning was wrong and has been corrected.** An SGLT2 inhibitor lowers the renal threshold but urinary glucose excretion still scales with plasma glucose above that threshold, so glycaemic improvement reduces UGE regardless — and from a higher baseline, by a larger absolute amount. Hall’s model accordingly predicts the *opposite* sign: stable background SGLT2i should produce the greatest attenuation. The preprint also distinguishes this from *co-initiation* of an SGLT2 inhibitor with an incretin agent, which it predicts should produce greater weight loss than either alone. Study 6 now tests all three strata.

Sources accessed at abstract level only (numbers limited to the abstract)

STEP 2, SURMOUNT-1, SCALE Obesity and Prediabetes, SCALE Diabetes, REDEFINE 1 and 2, ATTAIN-1 and 2, STEP-HFpEF and STEP-HFpEF DM (individually), STEP 6, OASIS 1, PIONEER PLUS, both retatrutide phase 2 trials, maridebart cafraglutide, Strathe 2023, and most physiology papers (Wing 1987, Bogardus 1986, Rave 2006, Heise 2022/2023, Ravussin 2025, Mäkimattila 1999, Carlson 1993).

Items not verified or unresolved

1. **STEP 2:** the value for the 1.0 mg arm and the baseline weight/BMI were not in the abstract and are not stated in the manuscript.
2. **Strathe 2023:** whether diabetes status or HbA1c reduced the modelled maximal weight response is not verified, because the full text was blocked by the publisher. The manuscript states this explicitly.
3. **STEP 6:** the subgroup by diabetes status is not in the abstract and was not used.
4. **Baseline characteristics (sex, age, background therapy) of several trials:** these are not reported in their abstracts, so between-programme composition differences are described only where verified (STEP 1, SURMOUNT-2, ATTAIN-2, maridebart cafraglutide, STEP-HFpEF pool).
5. **Glycogen–water contribution:** this is general-physiology inference without a cited human measurement in this context, and it is labelled as inference.
6. **Hypoglycaemia-avoidant eating** with sulfonylureas or insulin is presented as a hypothesis, without a supporting primary source.
7. **“No study has compared appetite suppression by diabetes status”**, **“no study measured muscle glucose uptake, oxidation, glycogen and total EE together”** and **“no study has measured endogenous incretin secretion, DPP-4 activity or autonomic function as predictors of weight response to an agonist”** are absence-of-evidence statements. They are limited by the scope of the search (PubMed only).
8. **Section 5 rests mainly on animal data.** Of the mechanisms in Section 5, only the secretion meta-analyses, the DPP-4 associations, the autonomic neuropathy prevalence, the gastric emptying distribution, the pharmacokinetics and the vagotomy study are human. The portal sensing, route-definition, neuropathy-function and agonist-response findings are all animal, in four different species, and none measured body weight.
9. **Several Section 2.5 sources were not number-verified.** The T2D-arm figures for naltrexone-bupropion [HOLLANDER2013] and lorcaserin [ONEIL2012] were verified against their abstracts. Their non-diabetes comparators [GREENWAY2010; SMITH2010], phentermine-topiramate [GADDE2011] and the bariatric cohort [COURCOULAS2015] are cited for the *direction* of the association only; no numbers from them appear in the text.
10. **SATTAR2025 has no abstract in its PubMed record** (it is a Letter). It is cited for the finding stated in its title; no numbers are taken from it.
11. **No human study has given a long-acting agonist to people differing in vagal function.** The central proposition of Section 5 is therefore untested in the population and with the drug class it concerns. Study 9 in Table 4 is the direct test.
12. **The receptor mediating hepatic vagal reception of portal GLP-1 has never been molecularly identified** [NISHIZAWA2000]. Statements about it are necessarily pharmacological inferences from antagonist and agonist insensitivity in anaesthetised rats.

Derived values and assumptions

- **Matched-pair ratios** (`analysis/derived.py`): these are ratios of published ETDs. Pairs are not independent (two doses come from the same trials), and no meta-analytic pooling or confidence intervals were computed for the ratios.
- **STEP 1 normoglycaemic estimate (~−15.9%):** this assumes the overall mean is the subgroup-size-weighted average of the subgroup means, and it ignores differences in imputation.
- **Energy scenarios (Table 2):** glycosuria reductions of 10/25/45 g/day and REE reductions of 0/40/90 kcal/day are assumptions anchored to measured studies in other settings. The conversion uses Hall 2011 (≈23.9 kcal/day per kg at steady state; exponential approach with a 1-year half-time). The scenarios are not predictions for any trial.

Analyses not performed

- Formal meta-analysis or meta-regression of the T2D × treatment interaction.
- Any quantitative apportionment of the weight-loss gap to peripheral versus central mechanisms. Section 5 deliberately reaches no numeric estimate, because no input to such a calculation has been measured in either population on treatment.
- Risk-of-bias assessment (RoB 2) and GRADE certainty rating.
- Adjustment of matched comparisons for age, sex, BMI or background therapy (requires IPD).
- Searches of Embase, Cochrane and regulatory documents (FDA/EMA reviews may contain subgroup analyses).

Access limitations

- Publisher sites (NEJM, Lancet, Wiley, figshare) refused automated access (HTTP 403 or bot challenge) even after network allowlisting.
- Supplementary appendices for most trials were therefore unavailable.
- Supplying the PDFs of STEP 2, SURMOUNT-1, REDEFINE 1/2, ATTAIN-1/2 and the STEP-HFpEF trials would allow verification of baseline composition and subgroup analyses (for example, weight response by baseline HbA1c or sulfonylurea use). This is the most valuable next step for strengthening Section 2.5.

Data and code availability

All derived values (matched-pair ratios, the implied normoglycaemic estimate and energy-balance scenarios) are generated by `analysis/derived.py`; figures by `figures/make_figures.py`; numbering and bibliography by `analysis/build_manuscript.py`. Bibliographic metadata were retrieved from PubMed (NCBI E-utilities).